

Loan-Eligibility&EMI-Predict-Ai

Loan Eligibility & Safe EMI Prediction System — Comprehensive Technical & Strategic Report

[INSERT YOUR NAME / CONSULTING TEAM NAME]

[INSERT ISSUE DATE]

Contents

About This Report	2
1. Executive Summary	4
2. Introduction	5
2.1 Background & Business Context	5
2.2 Problem Statement	6
2.3 Project Objectives & Scope	6
2.4 Stakeholder Map	7
2.5 How to Read This Report	7
3. Methodology	8
3.1 Analytical Framework	8
3.2 Data Governance & Statistical Validation	8
3.3 Model Development & Selection Methodology	9
3.4 Tools & Environment	10
4. Technical Architecture	10
4.1 System Architecture Overview	10
4.2 Data & Feature Engineering Pipeline	11
4.3 Model Layer	12
4.4 Application Layer	12
4.5 MLOps & Experiment Tracking	13
4.6 Technology Stack	13
5. Detailed Analysis	14
5.1 Dataset Profile	14
5.2 Exploratory Data Analysis	17
5.3 Statistical Hypothesis Testing	18
5.4 Feature Engineering Deep Dive	19
5.5 Classification Model Performance	20
5.6 Regression Model Performance	23
5.7 Model Explainability	24
5.8 Applicant Risk Segmentation	26
6. Market Analysis	27
6.1 Global AI-in-Lending Market Landscape	27
6.2 India Digital Lending Market Context	29

6.3 Competitive Landscape	29
6.4 Target Use Cases & Business Impact	30
7. Risk Assessment	30
7.1 Technical & Data Risk	30
7.2 Model Risk	31
7.3 Regulatory & Compliance Risk	32
7.4 Operational & Security Risk	33
7.5 Consolidated Risk Register	33
8. Future Scalability & Roadmap	34
8.1 Current-State Constraints	34
8.2 Scalability Roadmap	35
8.3 Phased Implementation Plan	36
9. Strategic Recommendations	37
10. Conclusion	38
Appendix A — Glossary of Terms	39
Appendix B — Full Metric Reference Tables	40
Appendix C — Data Dictionary	40
Appendix D — Sources & References	42
Appendix E — Visual Elements Guide	43
Appendix F — Formatting Instructions for Professional PDF Export	45

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About This Report

This report was developed directly from the EMIPredict AI codebase, notebook, and deployed application — not as a generic template. Every figure in Sections 1–5, 8, and the appendices (dataset size, model metrics, architecture, feature engineering logic) is drawn from the project’s own README, source code, trained model artifacts, and exploratory data analysis. Section 6 (Market Analysis) and parts of Section 7 (Risk Assessment) are grounded in current third-party market research and regulatory publications, cited inline and compiled in Appendix D.

Four fields remain as bracketed placeholders because they depend on information only your organization holds: **[PROJECT_GOALS]** (institution-specific KPI targets), **[KEY_STAKEHOLDERS]** (your governance and sign-off structure), and two narrower prompts inside the Technical Stack and Data Metrics tables where a deploying institution would typically insert its own infrastructure and volume figures alongside the project’s baseline numbers. Everywhere else, real project data has been used in place of a generic placeholder, since it produces a materially more useful document.

Manual Outline (see the auto-generated Contents page for linked, paginated navigation)

1. [Executive Summary](#)
2. [Introduction](#)
 - 2.1 Background & Business Context
 - 2.2 Problem Statement
 - 2.3 Project Objectives & Scope — **[PROJECT_GOALS]**
 - 2.4 Stakeholder Map — **[KEY_STAKEHOLDERS]**
 - 2.5 How to Read This Report
3. [Methodology](#)
 - 3.1 Analytical Framework
 - 3.2 Data Governance & Statistical Validation
 - 3.3 Model Development & Selection Methodology
 - 3.4 Tools & Environment
4. [Technical Architecture](#)
 - 4.1 System Architecture Overview
 - 4.2 Data & Feature Engineering Pipeline
 - 4.3 Model Layer
 - 4.4 Application Layer
 - 4.5 MLOps & Experiment Tracking
 - 4.6 Technology Stack — **[TECHNICAL_STACK]**
5. [Detailed Analysis](#)
 - 5.1 Dataset Profile — **[DATA_METRICS]**
 - 5.2 Exploratory Data Analysis
 - 5.3 Statistical Hypothesis Testing
 - 5.4 Feature Engineering Deep Dive
 - 5.5 Classification Model Performance
 - 5.6 Regression Model Performance
 - 5.7 Model Explainability
 - 5.8 Applicant Risk Segmentation
6. [Market Analysis](#)
 - 6.1 Global AI-in-Lending Market Landscape
 - 6.2 India Digital Lending Market Context
 - 6.3 Competitive Landscape
 - 6.4 Target Use Cases & Business Impact
7. [Risk Assessment](#)
 - 7.1 Technical & Data Risk
 - 7.2 Model Risk
 - 7.3 Regulatory & Compliance Risk
 - 7.4 Operational & Security Risk
 - 7.5 Consolidated Risk Register
8. [Future Scalability & Roadmap](#)
 - 8.1 Current-State Constraints
 - 8.2 Scalability Roadmap
 - 8.3 Phased Implementation Plan
9. [Strategic Recommendations](#)

10. Conclusion

11. Appendices

- Appendix A — Glossary of Terms
 - Appendix B — Full Metric Reference Tables
 - Appendix C — Data Dictionary
 - Appendix D — Sources & References
 - Appendix E — Visual Elements Guide (mapped to existing project assets)
 - Appendix F — Formatting Instructions for Professional PDF Export
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1. Executive Summary

EMIPredict AI is a completed, deployed machine learning platform that automates two linked lending decisions: whether a loan applicant is creditworthy (**eligibility classification**) and how large a monthly installment they can safely sustain (**maximum-EMI regression**). It was built end-to-end by a single AI/ML engineer, Mohit Singh Rajput, on a dataset of **404,800 historical loan applications**, and is publicly accessible today as a four-module Streamlit application.

The platform’s production model — XGBoost, selected competitively against three alternative algorithm families for each task — achieves **97.0% classification accuracy** (macro-F1 0.897, ROC-AUC 0.997) on a held-out test set for the three-class eligibility decision, and an **R² of 0.9935** (RMSE ₹625, MAE ₹229) on the safe-EMI regression task. These are strong, production-grade figures for a tabular financial risk-scoring problem, and they were arrived at through a disciplined process: systematic data cleaning, 42 engineered features (9 of them purpose-built financial ratios), three formally tested statistical hypotheses, and a four-model bake-off per task with hyperparameter search and MLflow experiment tracking.

This report evaluates EMIPredict AI as both a **technical asset** and a **strategic opportunity**. Structurally, it is organized around six questions a lending institution, investment committee, or engineering leadership team would need answered before relying on — or investing further in — a system like this:

Question	Where it’s answered
What does the system actually do, and how well?	Sections 4–5
Is there a real market for it?	Section 6
What could go wrong, and how exposed are we?	Section 7
What does it take to run this at institutional scale?	Section 8
What should we do next?	Section 9

Key findings:

- The modeling work is methodologically sound: XGBoost was not assumed but competitively selected against Logistic Regression, Decision Tree, and Random Forest baselines on both tasks, using macro-F1 and RMSE respectively, on a properly held-out 15% test split (70/15/15, seed-fixed for reproducibility).

- The three engineered affordability ratios — **EMI-to-income ratio**, **affordability ratio**, and **existing-loans flag** — account for the large majority of the classifier’s decision weight, which is exactly the intuitive, explainable behavior a credit risk model should exhibit and a point in favor of regulatory defensibility.
- The target classes are meaningfully imbalanced (77.3% Not_Eligible vs. 4.3% High_Risk), which the project addresses via class/sample weighting — an approach that should be revisited before any production deployment (see Section 7.2).
- The global AI-powered credit underwriting market is estimated at roughly **\$6-22 billion in 2026** depending on scope and research methodology, growing at a **17-19% CAGR** toward the mid-2030s, with India’s digital lending disbursement volume alone projected near **₹47.4 lakh crore (₹47.4 trillion) by 2026** (Section 6).
- The most material near-term risk is not model performance but **regulatory posture**: India’s RBI (Digital Lending) Directions, 2025 impose fairness-audit and algorithmic-disclosure obligations that this project, in its current portfolio/demo form, does not yet operationalize (Section 7.3).
- The system is currently a single-instance demo, not a production platform — it has no containerization, CI/CD, authentication, monitoring, or API layer. The project’s own README already identifies this gap; Section 8 turns that into a phased, sequenced roadmap.

Bottom line: EMIPredict AI demonstrates that the underlying prediction problem — translating income, expense, and credit data into a defensible eligibility decision and a safe EMI figure — is technically solvable to a high standard with a modest, well-engineered feature set. The gap between this project and an institution-grade lending decision engine is not a modeling gap; it is a **productionization, governance, and compliance gap**, and it is a closeable one. Section 9 sets out the recommended sequence for closing it.

2. Introduction

2.1 Background & Business Context

Consumer lending institutions — banks, non-banking financial companies (NBFCs), and digital-first fintech lenders — face a structural tension between speed and prudence. Applicants expect near-instant decisions; regulators and balance sheets demand consistent, defensible, low-default underwriting. Manual review does not scale to high application volumes without either slowing down or degrading in consistency across loan officers.

EMIPredict AI addresses a specific, well-scoped slice of that tension: given an applicant’s income, expenses, employment, credit history, and the loan they are requesting, (1) classify them into an eligibility tier, and (2) independently estimate the largest monthly installment they could safely carry — regardless of what they actually requested. The second output is what elevates this from a simple approve/reject classifier into an **affordability-based structuring tool**: an applicant who is “High_Risk”

for their requested amount may still be approvable at a lower, model-recommended installment.

The project was developed and documented as an individual capstone body of work in the Banking / Financial Services / Fintech Risk Assessment domain, following a structured, end-to-end data science lifecycle from raw data through a deployed, publicly accessible application.

2.2 Problem Statement

The project's own documentation frames the underlying business problem in five parts, all of which are common, well-recognized pain points in consumer lending operations:

1. Manual eligibility decisions are inconsistent across loan officers.
2. Lenders lack a systematic way to quantify how much EMI an applicant can actually afford, as distinct from what they requested.
3. Approving applicants who are statistically likely to default carries direct credit-loss risk.
4. Manual underwriting does not scale efficiently to high application volumes.
5. There is no consistent, repeatable method for translating raw income, expense, and credit-history fields into a lending decision.

EMIPredict AI's dual-model design — a classifier for the eligibility decision and a regressor for the affordability ceiling — is a direct response to problems 2 and 5 in particular: it does not just gate applicants in or out, it produces a number the applicant and the institution can negotiate around.

2.3 Project Objectives & Scope

As documented in the project itself, the objectives are:

Eligibility classification - Analyze applicant income, expense, credit, and loan-request data. - Engineer affordability and debt-burden ratios from raw fields. - Classify applicants into Eligible, High_Risk, or Not_Eligible. - Flag high-risk applicants ahead of disbursal rather than after default.

Safe EMI estimation - Build a regression model for the maximum sustainable monthly EMI. - Compare the model's recommended EMI against the amount actually requested. - Support affordability-based loan structuring in place of a flat approve/reject gate.

Institution-specific objectives:

[PROJECT_GOALS] — Insert the specific, measurable targets your organization is adopting this system to hit. Typical examples in this domain include: target reduction in average underwriting turnaround time (e.g., “reduce first-pass decision time from 3 business days to under 5 minutes”); target improvement in portfolio default/NPA rate; target loan-officer throughput gain; target reduction in over-lending incidents; or a target increase in

approval rate among currently-declined-but-creditworthy applicants (financial inclusion framing — see Section 6.1 for supporting market evidence on this point). These targets should be set by risk, credit policy, and product stakeholders before development scope is finalized, and they materially affect which recommendations in Section 9 apply to your situation.

Scope of this report: this document evaluates EMIPredict AI as it exists today — a validated modeling pipeline behind a demonstration Streamlit application — and assesses what is required to move it toward an institution-grade production capability. It does not assume access to any specific institution’s loan book, existing risk infrastructure, or regulatory approvals; those are called out as placeholders wherever they would change the analysis.

2.4 Stakeholder Map

[KEY_STAKEHOLDERS]

No organizational stakeholder map exists in the source project — its only listed stakeholder is its individual author/engineer. Before this system is taken toward production, the following roles typically need to be identified by name and engaged at the review gates noted in Section 8.3. Insert your organization’s actual owners in the table below; the role list is a standard starting structure for a lending-decision AI system, not a prescriptive requirement.

Role	Responsibility	Name / Owner
Executive Sponsor	Budget authority, strategic alignment	[INSERT]
Head of Credit Risk / Chief Risk Officer	Approves eligibility/EMI policy thresholds	[INSERT]
Compliance & Regulatory Affairs Lead	RBI / data-protection sign-off (Section 7.3)	[INSERT]
Data Protection Officer	DPDP Act 2023 compliance, consent handling	[INSERT]
Engineering / MLOps Lead	Productionization roadmap owner (Section 8)	[INSERT]
Model Risk / Validation Function	Independent model review and bias testing	[INSERT]
Product Owner	Loan-officer and applicant-facing UX	[INSERT]
External Auditor (if applicable)	Periodic algorithmic audit per RBI guidance	[INSERT]

2.5 How to Read This Report

Readers with a **technical** focus should prioritize Sections 3–5 and Appendix C. Readers with a **business or investment** focus should prioritize Sections 1, 6, and 9.

Readers with a **risk, compliance, or governance** mandate should prioritize Sections 7 and 8.3. Every metric quoted in Sections 3–5 is reproducible directly from the project’s own notebook (notebooks/EMIPredict_AI_Model_Development.ipynb) and is cross-referenced to the specific model artifact or chart that produced it.

3. Methodology

3.1 Analytical Framework

The project follows a linear, auditable analytical lifecycle, functionally equivalent to the industry-standard CRISP-DM cycle (Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment), and it is explicit and reproducible at every stage. The project’s own architecture diagram renders it as:

Raw Applicant Data → Data Cleaning → EDA → Financial Ratio Feature Engineering
→ Feature Scaling & Encoding → Classification + Regression Models
→ Model Selection (best of 4 candidates per task) → MLflow Tracking
→ Streamlit Deployment

Two properties of this pipeline are worth calling out explicitly because they are not universal in portfolio-grade ML projects and materially affect the trustworthiness of the results in Section 5:

1. **A genuine held-out test set.** The 70/15/15 train/validation/test split is applied once, with a fixed random seed (`random_state = 42`), and every metric reported in this document is computed on the 15% test partition the models never saw during training or hyperparameter tuning — not on training or validation data. This matters because a model’s training-set accuracy is not evidence of anything; its held-out test performance is.
2. **A like-for-like, four-model bake-off per task**, not a single model trained and reported. Section 5.5 and 5.6 present the full comparison, not just the winner, which is what allows an outside reviewer to judge how much incremental accuracy XGBoost is actually buying versus a simpler, more interpretable Decision Tree or Random Forest.

3.2 Data Governance & Statistical Validation

Before any modeling, the raw dataset was put through a documented data-quality pass. This is a frequently-skipped step in portfolio ML work, and its presence here is a positive signal for data governance maturity:

Issue identified	Remediation applied
Corrupted numeric strings in <code>age</code> , <code>monthly_salary</code> , <code>bank_balance</code>	Regex-based cleaning, re-cast to numeric
Inconsistent gender labels	Standardized to Male / Female

Issue identified	Remediation applied
Out-of-range credit scores (outside the valid 300-850 band)	Set to missing, then imputed
Missing education	Imputed with mode
Missing monthly_rent	Imputed with median, grouped by house_type
Missing credit_score, bank_balance, emergency_fund	Imputed with column median
family_size collinear with dependents (r = 1.00)	Dropped from the feature set

Beyond descriptive cleaning, the project applies **formal statistical hypothesis testing** — at a 5% significance level ($\alpha = 0.05$) — to confirm that three relationships suggested by the exploratory charts are real rather than visual artifacts. This is a meaningfully more rigorous step than most comparable projects take, and its results are detailed in Section 5.3. Critically, the project’s own analysis explicitly distinguishes **statistical significance** from **practical effect size** — noting that with a sample of 404,800 rows, even small, practically unimportant differences will register as statistically significant, and that the effect size (not just the p-value) is what should inform business decisions. That distinction is good statistical practice and is called out here because it directly informs how Section 5.3’s findings should be weighted.

3.3 Model Development & Selection Methodology

Both prediction tasks — classification and regression — were approached with an identical structure: a linear/baseline model, two tree-based ensemble alternatives, and a gradient-boosted final candidate.

Step	Classification track	Regression track
Baseline	Logistic Regression (class-balanced)	Linear Regression
Alternative 1	Decision Tree (GridSearchCV)	Decision Tree Regressor (GridSearchCV)
Alternative 2	Random Forest (GridSearchCV)	Random Forest Regressor (GridSearchCV, subsampled)
Final candidate	XGBoost (RandomizedSearchCV, sample-weighted)	XGBoost Regressor (RandomizedSearchCV)
Selection metric	Test-set macro-F1	Test-set RMSE

GridSearchCV (exhaustive search) was used for the three simpler model families, where the hyperparameter space is small enough to search completely; XGBoost, with a larger and more continuous hyperparameter space (`n_estimators`, `max_depth`, `learning_rate`, `subsample`), was tuned with `RandomizedSearchCV`, which samples the

space rather than enumerating it — the standard, appropriate trade-off for a gradient-boosted model of this size. Class imbalance in the eligibility target (Section 5.1) was handled via class-weighted and sample-weighted training rather than resampling (e.g., SMOTE), which avoids introducing synthetic data into a financial-risk training set — a defensible and generally preferable choice in a regulated-lending context, since synthetic oversampling can be harder to justify to a model-risk reviewer or regulator than a weighting scheme applied to real observations.

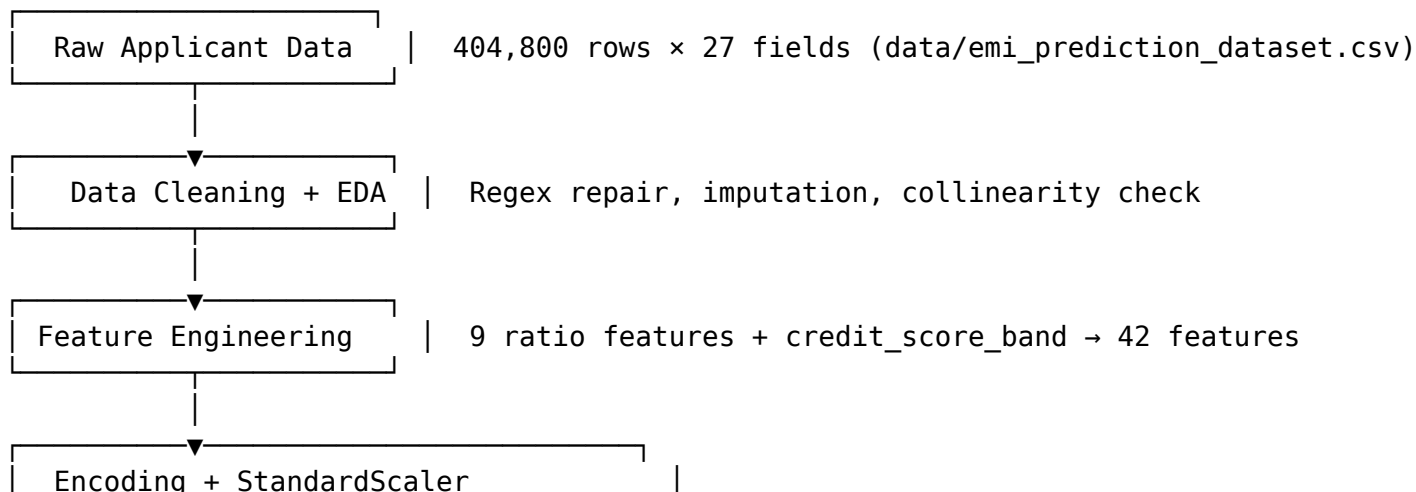
3.4 Tools & Environment

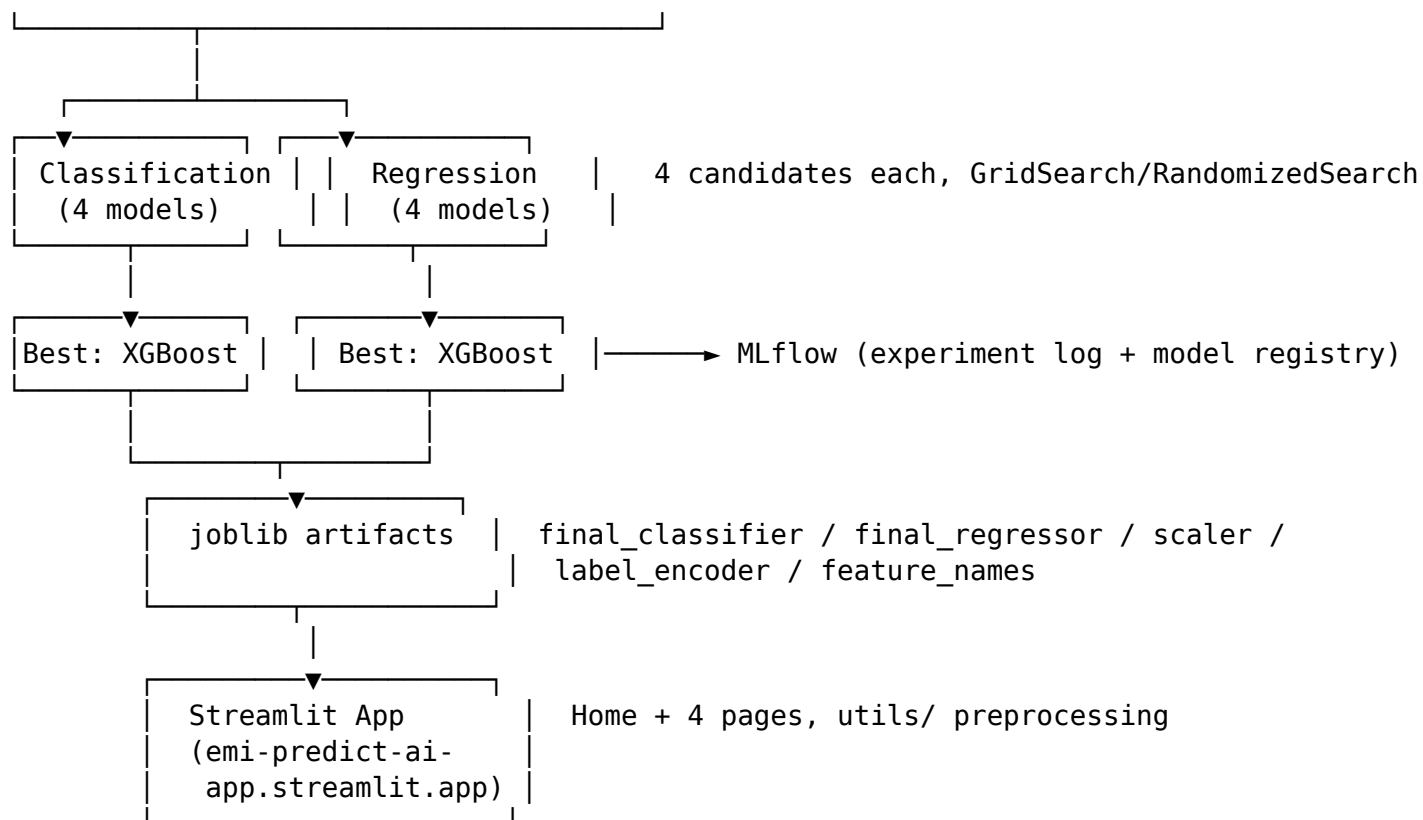
The full technical environment is detailed in Section 4.6; at the methodology level, three environment choices are worth noting because they affect reproducibility and auditability: the random seed is fixed and stated (`random_state = 42`) throughout data splitting and model training; every model run is logged to **MLflow**, giving a queryable experiment history rather than only the final artifact; and the repository ships a `.devcontainer` configuration, meaning the development environment itself (not just the code) is reproducible via VS Code / GitHub Codespaces — a detail that matters when a validation team or auditor needs to re-run the notebook end-to-end rather than trust a static report.

4. Technical Architecture

4.1 System Architecture Overview

EMIPredict AI is organized into four architectural layers: a **data layer** (the raw and engineered applicant dataset), a **model layer** (trained, serialized classifier and regressor plus their supporting preprocessing artifacts), an **application layer** (a multi-page Streamlit web app), and an **experiment/tracking layer** (MLflow, sitting alongside — not inside — the request path). The project’s own system diagram (rendered in Mermaid in the source README) captures the training-time flow; the diagram below extends it to show the artifact handoff into the application layer, which is the part most relevant to a production readiness review:





4.2 Data & Feature Engineering Pipeline

The raw dataset carries 27 fields per applicant — demographic (age, gender, marital status, education, dependents), employment (type, tenure, company type), housing (type, rent), income and existing obligations (salary, current EMI, existing loans), liquidity (bank balance, emergency fund), credit history (credit score), and the requested loan itself (scenario, amount, tenure). Section 5.4 details the transformation of these 27 fields into the 42-feature model input; architecturally, the important point is that **the exact same feature-construction logic runs twice** — once in the training notebook, and once at inference time inside `utils/preprocessing.py`, whose `build_feature_row()` function reconstructs every engineered ratio (debt-to-income, affordability ratio, disposable income, EMI-to-income ratio, liquidity ratio, emergency-fund-months, and the credit-score band) from a single applicant’s raw form input, then column-orders the result to match the `feature_names`. joblib artifact the models were trained on. Keeping training-time and inference-time feature logic in the same numerical form (not just conceptually equivalent, but literally the same formulas) is one of the more common sources of silent production ML bugs — training/serving skew — and this project’s design explicitly guards against it by centralizing the formulas in one module with a code comment tying it back to the notebook section that defines the canonical version.

4.3 Model Layer

Two models are in production, both XGBoost, each paired with the preprocessing artifacts needed to serve it:

Artifact	File	Purpose
Classifier	models/final_classifier.joblib (4.8 MB)	0-class eligibility prediction
Regressor	models/final_regressor.joblib (1.6 MB)	Max safe monthly EMI (₹)
Scaler	models/scaler.joblib	StandardScaler, applied only ahead of the linear baselines
Label encoder	models/label_encoder.joblib	Maps encoded classes back to Eligible / High_Risk / Not_Eligible
Feature order	models/feature_names.joblib	Canonical 42-feature column order expected by both models

A notable, correct implementation detail: the application checks `needs_scaling()` against the model's class name before deciding whether to apply the saved `StandardScaler` to an inference request. Because the deployed models are both XGBoost — a tree-based algorithm that does not require feature scaling — the scaler is retained in the artifact set (for the linear baselines, and for any future retraining that reintroduces them) but is correctly bypassed on the production inference path. This is a small but telling sign of care in the engineering, not just the modeling.

4.4 Application Layer

The user-facing system is a five-view Streamlit application:

View	File	Function
Home	Home.py	Landing page; model snapshot cards (algorithm, accuracy, R^2)
Eligibility Prediction	pages/1_Eligibility_Prediction.py	Runs the classifier; shows predicted class + full probability breakdown
EMI Calculator	pages/2_EMI_Calculator.py	Runs the regressor; compares safe EMI vs. requested installment
Model Performance	pages/3_Model_Performance.py	Renders the full 4-model comparison, confusion matrix, feature importance

View	File	Function
Data Insights	pages/4_Data_Insights.py	Dataset-level trends; live, user-filterable segment explorer

Two engineering choices are worth flagging as strengths for a production readiness review. First, model and dataset loading are wrapped in Streamlit's `@st.cache_resource` and `@st.cache_data` decorators respectively (in `utils/model_loader.py`), so the ~6.4 MB of joblib artifacts and a 20,000-row dataset sample are read from disk once per session rather than on every page interaction — a meaningful responsiveness decision at this artifact size. Second, the Eligibility Prediction page surfaces the **full class-probability distribution**, not just the arg-max predicted label — this is the right default for a lending decision surface, since it lets a loan officer see how close a “High_Risk” call was to “Eligible” rather than presenting a false binary.

4.5 MLOps & Experiment Tracking

MLflow is used to log every training run across both tasks and register the best-performing model per task. This gives the project queryable experiment history — parameters, metrics, and artifacts per run — beyond just the two final `.joblib` files, which is the foundation a model-risk or validation function would need to reconstruct *why* XGBoost was selected, not just *that* it was. Section 7.1 and 8.2 discuss what is still missing to call this a full MLOps practice (model versioning in a registry with promotion gates, automated retraining triggers, drift monitoring) versus experiment tracking, which is what is implemented today.

4.6 Technology Stack

[TECHNICAL_STACK]

The stack below is the project's actual, verified dependency set (`requirements.txt`), not a generic recommendation. If your organization is adopting or extending this system, add your own infrastructure choices — cloud provider, container orchestration, CI/CD platform, API gateway, and identity provider — in the row marked accordingly; none of those currently exist in the source project (see Section 8).

Layer	Technology	Notes
Language	Python 3.10+	
Data processing	Pandas $\geq$ 2.1, NumPy $\geq$ 1.26	
Statistics	SciPy $\geq$ 1.11	Powers the hypothesis tests in Section 5.3
Visualization	Matplotlib $\geq$ 3.8, Seaborn $\geq$ 0.13	All 14+ EDA and evaluation charts

Layer	Technology	Notes
Classical ML	Scikit-learn $\geq$ 1.3	Logistic/Linear Regression, Decision Tree, Random Forest, StandardScaler, GridSearchCV
Gradient boosting	XGBoost $\geq$ 2.0	Final production classifier and regressor
Experiment tracking	MLflow $\geq$ 2.9	Run logging, model registry
Serialization	Joblib $\geq$ 1.3	Model and preprocessing artifact persistence
Application framework	Streamlit $\geq$ 1.29	Multi-page web application
Notebook environment	Jupyter Notebook	Model development (notebooks/EMIPredict_AI_Model_Devel
Dev environment	.devcontainer (VS Code / Codespaces)	Reproducible development container
Current deployment target	Streamlit Community Cloud	Free tier; sleeps after inactivity — see Section 7.4
Production additions (institution-specific)	[INSERT — e.g., AWS/Azure/GCP, Docker, Kubernetes, GitHub Actions, API gateway, IdP]	Not present in source project; see Section 8.2

5. Detailed Analysis

5.1 Dataset Profile

[DATA_METRICS]

All figures below were independently verified against the raw dataset (data/emi_prediction_dataset) for this report, not taken only from summary documentation.

Metric	Value
Total applicant records	404,800
Raw fields	27
Engineered feature count (model input)	42
Classification target	emi_eligibility (3 classes)
Regression target	max_monthly_emi (₹, continuous)
Train / Validation / Test split	70% / 15% / 15%
Random seed	42

Eligibility class distribution (verified row counts):

Class	Count	Share
Not_Eligible	312,868	77.29%
Eligible	74,444	18.39%
High_Risk	17,488	4.32%

The majority-to-minority class ratio is approximately **17.9 : 1** (Not_Eligible : High_Risk). This is a real, material imbalance — not a modeling inconvenience to wave away — and it is the single most important statistic to carry into Section 7.2’s discussion of model risk, because it directly affects how much confidence the macro-averaged metrics in Section 5.5 deserve.

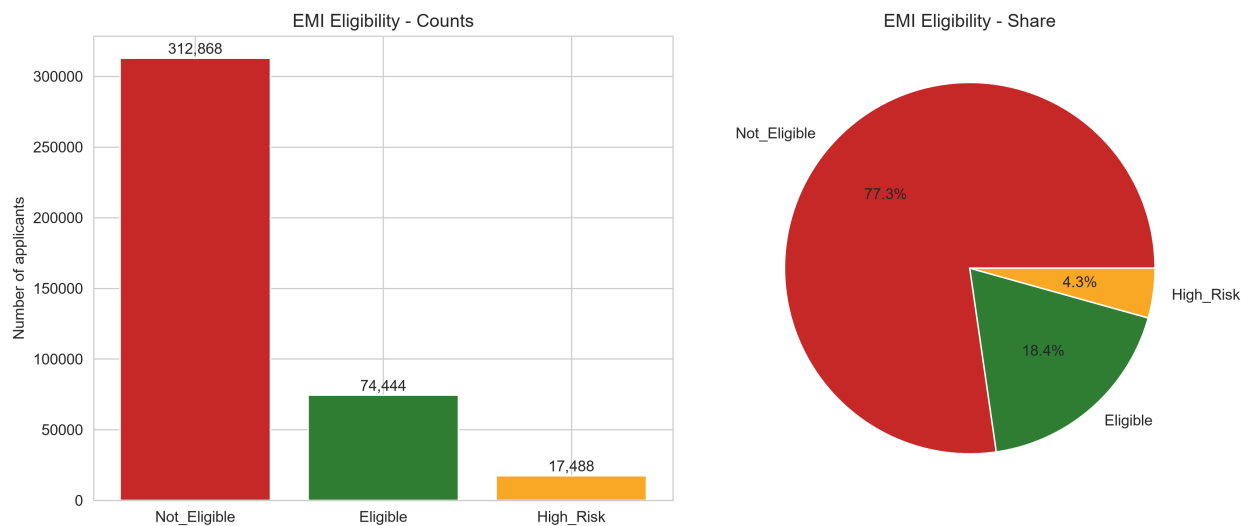


Figure 5.1 — Eligibility class distribution across all 404,800 applicants. Source: EMIPredict AI model-development notebook.

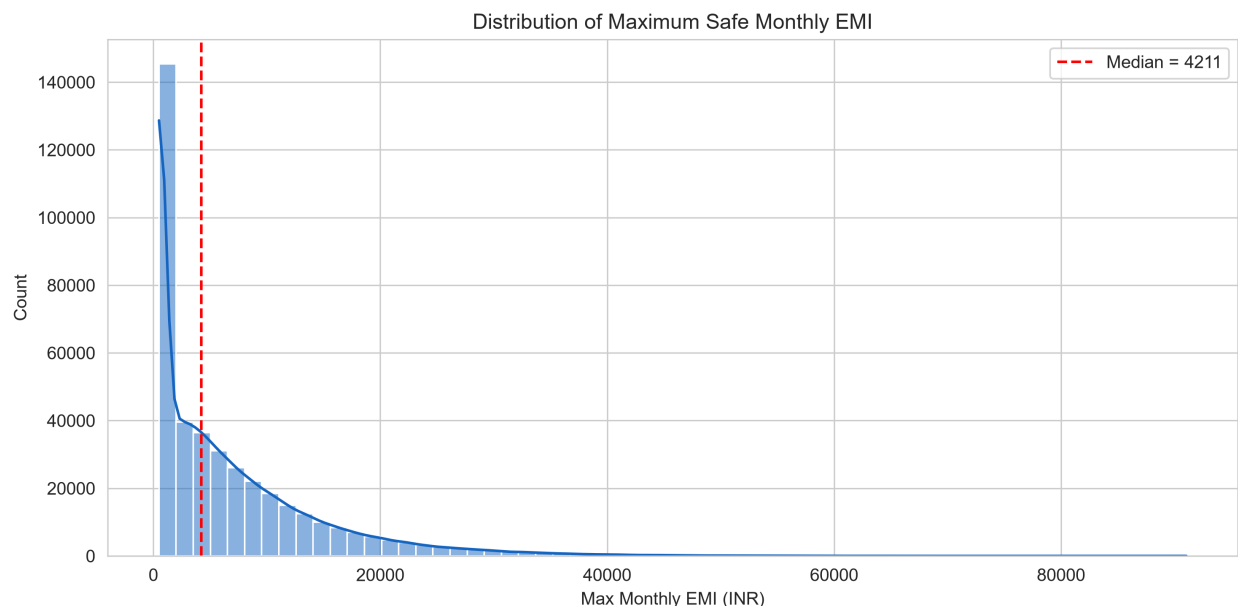


Figure 5.1b — Distribution of the regression target, `max_monthly_emi`, showing the right-skew discussed in Section 5.2. Source: EMIPredict AI model-development notebook.

Applicant population profile (verified row counts):

Dimension	Breakdown
Employment type	Private 283,099 (69.9%) · Government 81,167 (20.1%) · Self-employed 40,534 (10.0%)
Employer type	Large Indian 121,139 (29.9%) · MNC 101,409 (25.1%) · Mid-size 101,301 (25.0%) · Startup 60,706 (15.0%) · Small 20,245 (5.0%)
Housing	Rented 161,601 (39.9%) · Own 142,307 (35.2%) · Family 100,892 (24.9%)
Loan purpose (<code>emi_scenario</code>)	Near-evenly split across five scenarios — Home Appliances, Personal Loan, E-commerce Shopping, Education, Vehicle EMI — each 80,900-81,000 records (~20% apiece)

[INSERT — deploying institutions should benchmark their own applicant population against this profile before deploying the model as-is; a portfolio with a materially different employment, income, or housing mix than the one above is effectively out-of-distribution for this model and would need a validation sample and, likely, retraining before go-live — see Section 7.2.]

5.2 Exploratory Data Analysis

The project’s EDA follows a structured univariate → bivariate → multivariate progression across 14 charts, each interrogated for “why this chart,” “what it shows,” “the positive business implication,” and “the risk implication” — a discipline that keeps the analysis tied to lending decisions rather than generic data description. The headline findings:

- **Affordability, not identity, drives eligibility.** Debt-to-income and EMI-to-income ratios show the strongest relationship with eligibility outcomes of any variables tested — stronger than any single demographic field. This is echoed quantitatively in Section 5.7’s feature-importance results and is a meaningfully positive finding from a fair-lending perspective (Section 7.3): the model’s strongest signal is a financial-capacity measure, not a demographic proxy.
- **max_monthly_emi is right-skewed** — consistent with a small share of higher-income applicants requesting (and being approved for) disproportionately large EMIs, as would be expected in a real income distribution.
- **Credit score separates the three eligibility classes with a real, if moderate, gap** — median credit score rises from 695 (Not_Eligible) to 714 (High_Risk) to 725 (Eligible). The gap is statistically confirmed (Section 5.3) but the boxes overlap substantially, meaning credit score alone is a weak classifier — consistent with the feature-importance ranking in Section 5.7, where credit score sits mid-table rather than at the top.
- **family_size and dependents are perfectly collinear** ($r = 1.00$), confirmed both visually via the correlation heatmap and numerically; only dependents was retained.

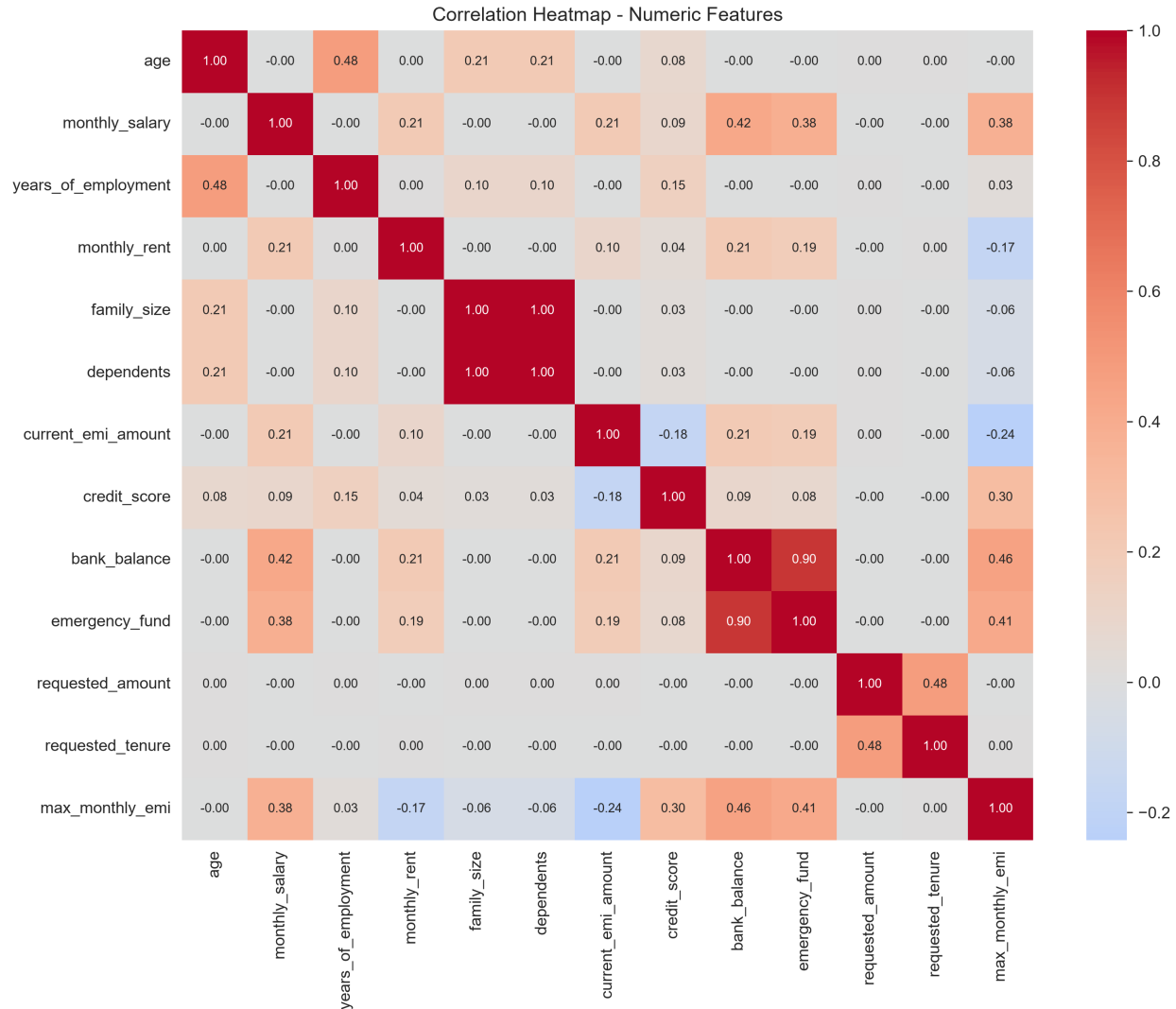


Figure 5.2 — Full numeric feature correlation matrix, including the *family_size* / *dependents* collinearity finding. Source: EMIPredict AI model-development notebook.

- **Employment type shows only a modest practical association with eligibility** despite reaching statistical significance (Section 5.3) — no single employment category (Government, Private, Self-employed) stands out as dramatically safer or riskier once affordability ratios are accounted for.

5.3 Statistical Hypothesis Testing

Three relationships suggested by the EDA charts were formally tested at $\alpha = 0.05$ rather than read qualitatively off a plot. All three null hypotheses were rejected; the table below adds the effect-size context needed to interpret that correctly.

#	Test	Null Hypothesis	Result	Effect Size / Practical Read
1	One-way ANOVA	Mean credit score is equal across Eligible / High_Risk / Not_Eligible	Rejected ($p < 0.001$)	Real but modest — median gap of 30 points (695 → 725) across three classes
2	Pearson correlation	No linear correlation between monthly salary and max safe EMI	Rejected ($p < 0.001$)	Moderate — $r \approx 0.38$
3	Chi-square test of independence	Employment type and eligibility are independent	Rejected ($p < 0.001$)	Statistically real, practically modest — driven partly by sample size

The project’s own analysis flags test #3 as the clearest illustration of a principle every reader of this report’s Section 6 and 9 should hold onto: **at $n = 404,800$, statistical significance is cheap**. Almost any real-world association will clear $p < 0.05$ at this sample size, including ones too small to act on. The right question for a business or compliance stakeholder is never “was it significant?” — it is “how large is the effect, and does it justify a policy or pricing decision?” This project asks that second question explicitly, which is a meaningfully more mature analytical posture than treating $p < 0.05$ alone as a green light — and it is the standard this report holds Section 6’s market-sizing claims to as well.

5.4 Feature Engineering Deep Dive

The transformation from 27 raw fields to 42 model-ready features is where most of this project’s domain expertise is expressed. Nine engineered financial ratios do the heaviest lifting:

Engineered Feature	Formula
total_monthly_expenses	Sum of rent, school/college fees, travel, groceries/utilities, other expenses
debt_to_income_ratio	$\text{current_emi_amount} \div \text{monthly_salary}$
expense_to_income_ratio	$\text{total_monthly_expenses} \div \text{monthly_salary}$

Engineered Feature	Formula
<code>disposable_income</code>	<code>monthly_salary - total_monthly_expenses - current_emi_amount</code>
<code>affordability_ratio</code>	<code>disposable_income ÷ monthly_salary</code>
<code>proposed_installment</code>	<code>requested_amount ÷ requested_tenure</code>
<code>emi_to_income_ratio</code>	<code>proposed_installment ÷ monthly_salary</code>
<code>emergency_fund_months</code>	<code>emergency_fund ÷ total_monthly_expenses</code>
<code>liquidity_ratio</code>	<code>bank_balance ÷ total_monthly_expenses</code>

A tenth engineered feature, `credit_score_band`, bins the continuous credit score into five ordinal tiers (Poor / Fair / Good / Very_Good / Excellent, using bin edges at 579 / 669 / 739 / 799 / 850) — giving the models both the raw continuous score and a business-interpretable band, which is useful when explaining a decision to an applicant or auditor in policy language rather than a raw numeric threshold.

Categorical encoding follows standard, defensible practice: binary fields (gender, marital status, existing loans) are label-mapped to 0/1; ordinal fields (education, credit score band) preserve their natural order; and nominal fields with no inherent order (employment type, employer type, housing type, loan scenario) are one-hot encoded with one category dropped per field as the implicit reference level — correctly avoiding the dummy-variable trap. This pattern is applied consistently, including at inference time in `utils/preprocessing.py`.

One implementation observation worth flagging for a production hardening pass (carried forward into Section 7.1): the one-hot category lists in `utils/preprocessing.py` are hardcoded Python lists rather than being derived programmatically from a saved category vocabulary. This works correctly today because it mirrors the training-time encoding exactly, but it creates a maintainability risk — if the underlying applicant population later includes a new loan scenario, employer type, or housing category not present in the original 404,800-row training set, the current code will silently encode it as the reference category rather than raising a validation error, which could produce a misleadingly confident prediction on genuinely out-of-distribution input. Deriving the expected category set from `feature_names.joblib` (or a saved vocabulary artifact) with an explicit “unseen category” guard is a small, worthwhile fix ahead of production use.

5.5 Classification Model Performance

Four algorithms were trained and evaluated on the identical held-out test split for the eligibility classification task:

Rank	Model	Accuracy	F1 (macro)	Notes
1	XGBoost	97.0%	0.897	Final production model
2	Decision Tree	93.3%	0.800	Strong, more interpretable alternative
3	Random Forest	92.3%	0.793	Comparable to Decision Tree
4	Logistic Regression	81.9%	0.669	Linear baseline

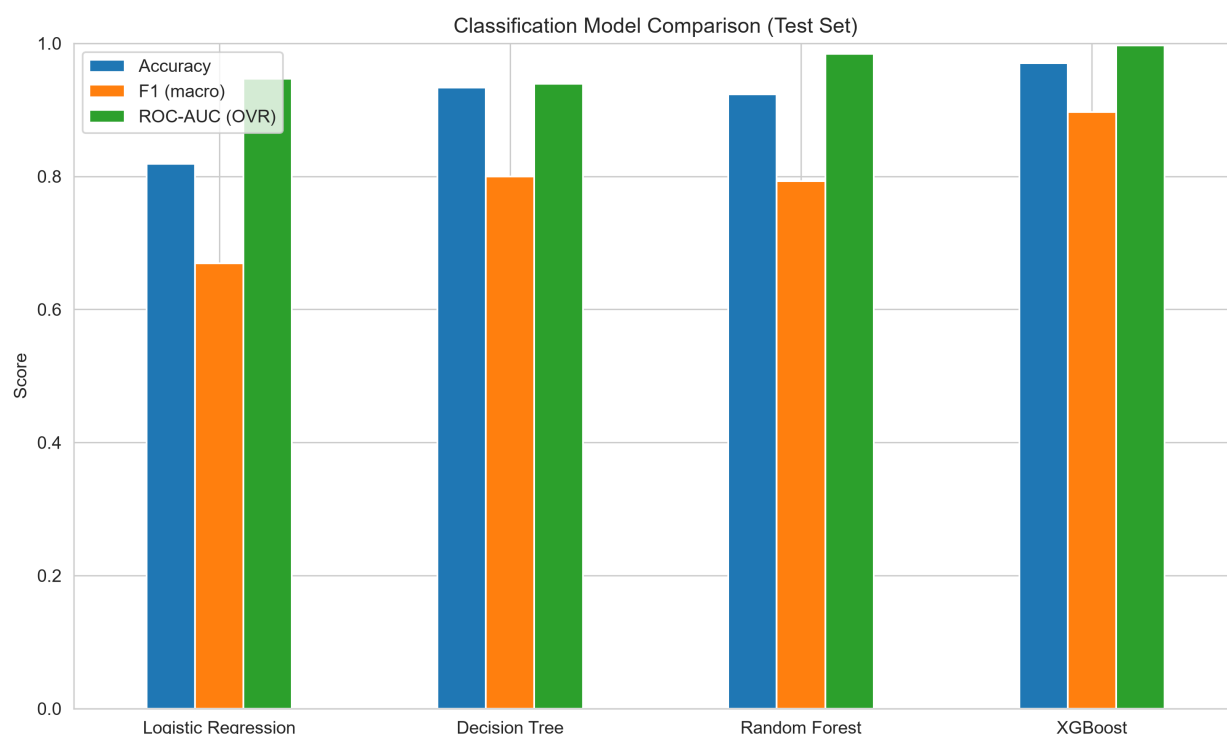


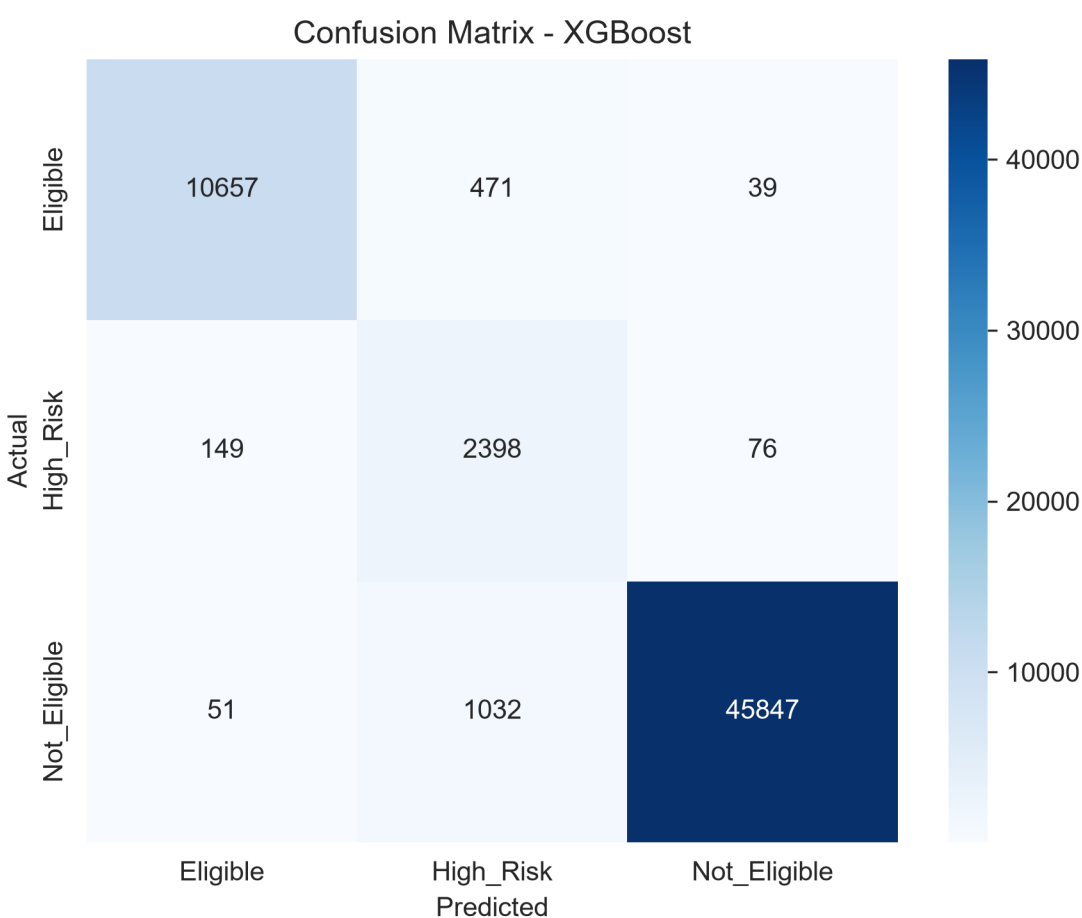
Figure 5.5a — Accuracy, macro-F1, and ROC-AUC across all four classifiers. Source: EMIPredict AI model-development notebook.

Final model (XGBoost) test-set metrics:

Metric	Score
Accuracy	97.0%
Precision (macro)	0.865
Recall (macro)	0.949
F1-score (macro)	0.897
ROC-AUC (one-vs-rest)	0.997

Two things are worth an outside reader's attention here, beyond the headline 97.0%

accuracy. First, **macro-averaged precision (0.865) trails macro-averaged recall (0.949)** — meaning the model is somewhat more prone to false positives across classes (predicting a risk tier the applicant doesn't truly belong to) than to missing true instances of a class. In a lending context this is broadly the safer direction to err in for the risk-averse classes, but it should be validated per-class (not just macro-averaged) against the institution's specific cost-of-false-approval versus cost-of-false-decline before setting decision thresholds — the macro metrics reported here and in the source project do not, by themselves, tell you the per-class confusion pattern for the minority `High_Risk` class, which is the class a risk committee will care about most. Second, the **17.9:1 class imbalance noted in Section 5.1** means accuracy alone is a weak headline metric — a model that always predicted `Not_Eligible` would already score 77.3% accuracy while being useless. The fact that XGBoost's macro-F1 (0.897, which weights all three classes equally regardless of size) and ROC-AUC (0.997) are both strong is the more meaningful evidence of genuine predictive skill across all three classes, not just the majority one.



Figure

5.5b — XGBoost classifier confusion matrix on the held-out test set — the direct visual for Recommendation 1 in Section 9. Source: EMIPredict AI model-development notebook.

5.6 Regression Model Performance

The same four-model structure was applied to the safe-EMI regression task:

Rank	Model	RMSE	R ²	Notes
1	XGBoost	₹625	0.994	Final production model
2	Random Forest	₹1,086	0.980	Strong alternative
3	Decision Tree	₹1,155	0.978	Comparable to Random Forest
4	Linear Regression	₹4,100	0.720	Linear baseline

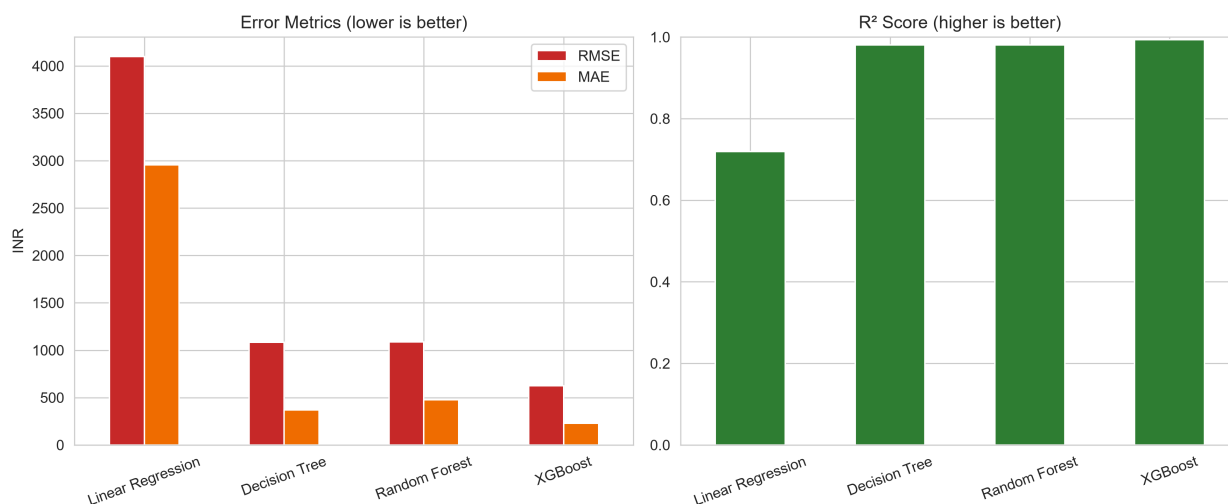


Figure 5.6a — RMSE and R² across all four regressors. Source: EMIPredict AI model-development notebook.

Final model (XGBoost) test-set metrics:

Metric	Score
RMSE	₹625
MAE	₹229
R ²	0.9935

The gap between the linear baseline (R² 0.720) and the tree-based models (R² ≥ 0.978) is unusually large for this kind of tabular financial data, and it is informative: it indicates the true relationship between an applicant’s financial profile and their safe EMI ceiling is meaningfully **non-linear** — most plausibly driven by threshold and interaction effects (e.g., affordability ratio behaving differently once disposable income crosses certain bands, or the interaction between existing debt load and requested tenure) that a single linear model structurally cannot capture, but that tree-based splits can. This is a genuinely useful finding independent of which specific algorithm is deployed: it says something about the shape of affordability itself, and it is a reason to be cautious about any future simplification of the production model back toward a linear or rules-based system for interpretability reasons (Section 7.3) — doing so would trade away real predictive accuracy, not just model complexity.

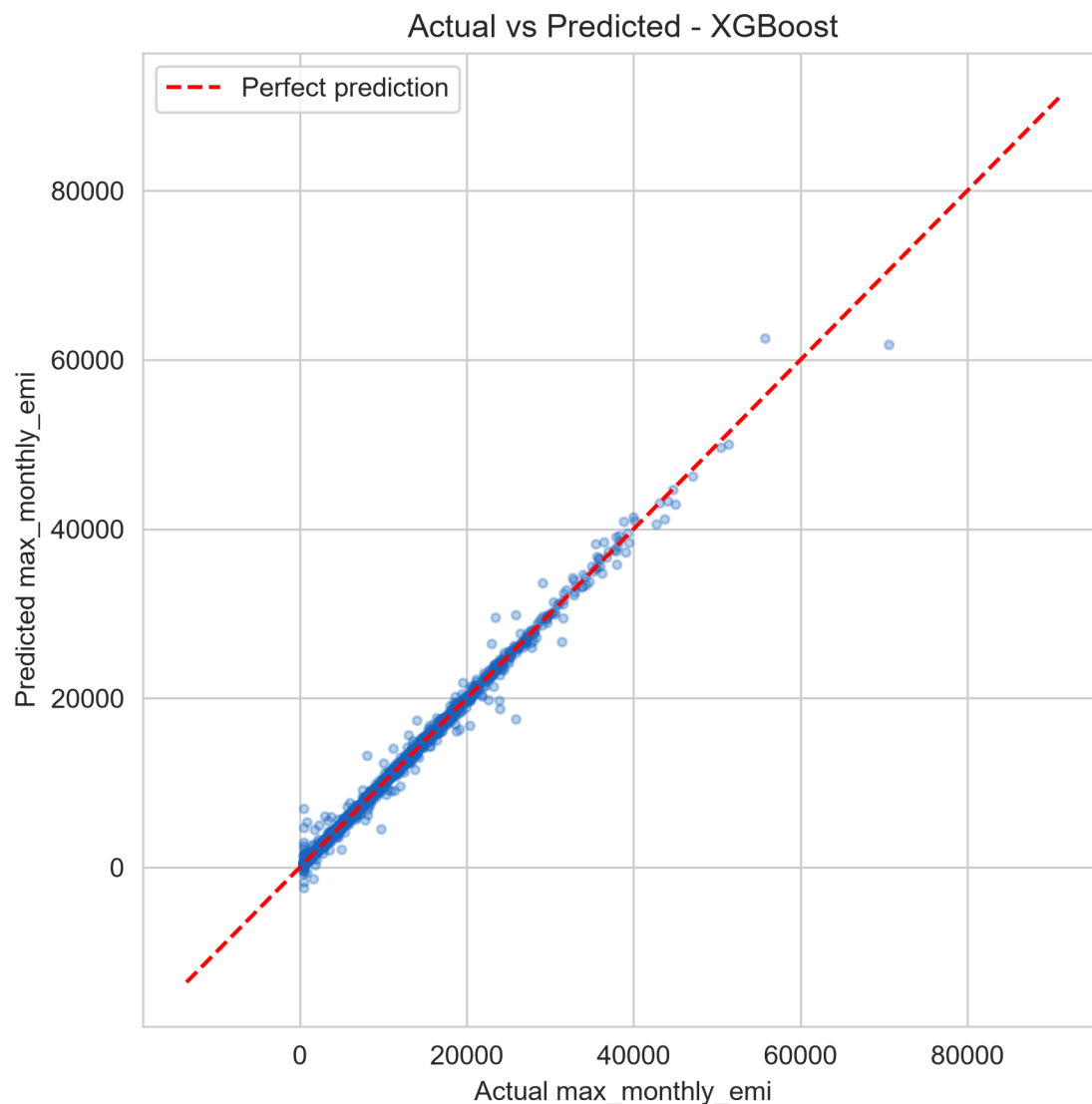


Figure 5.6b — Actual vs. predicted maximum safe EMI on the test set ($R^2 = 0.9935$). Source: EMIPredict AI model-development notebook.

5.7 Model Explainability

The XGBoost classifier’s feature importance ranking is the single most important exhibit in this report for a compliance or fair-lending review, because it answers the question “what is this model actually deciding on?” The top-ranked features, in descending order of importance, are:

1. **emi_to_income_ratio** — by a clear margin the single most influential feature
2. **affordability_ratio** — close second
3. **existing_loans** — close third
4. **disposable_income**
5. **credit_score**
6. **current_emi_amount**
7. **debt_to_income_ratio**

followed by a longer tail including loan scenario, monthly rent, employment type, monthly salary, housing type, liquidity ratio, and proposed installment, each contributing a smaller, roughly comparable share.

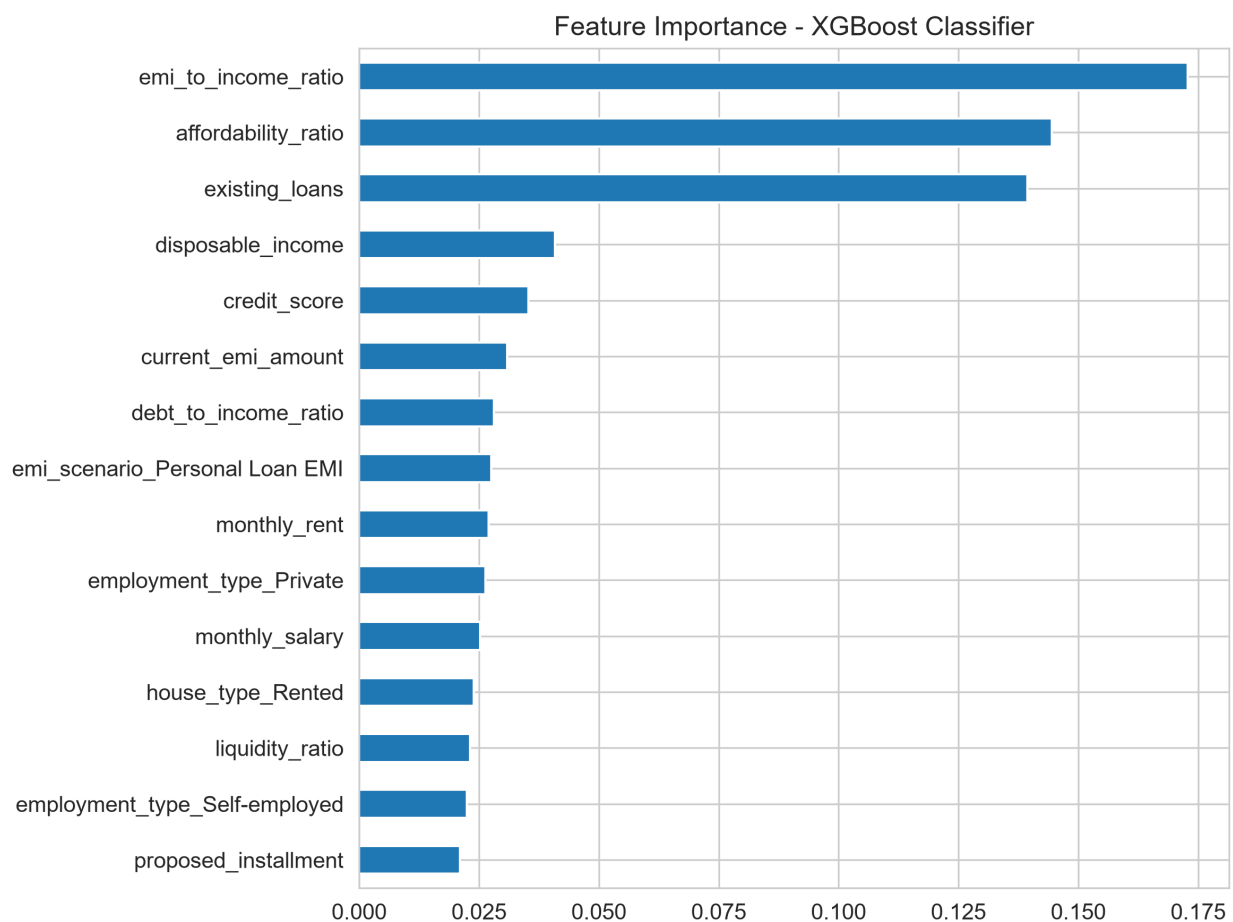


Figure 5.7a — XGBoost classifier feature importance, the chart underlying all of Section 5.7's findings. Source: EMIPredict AI model-development notebook.

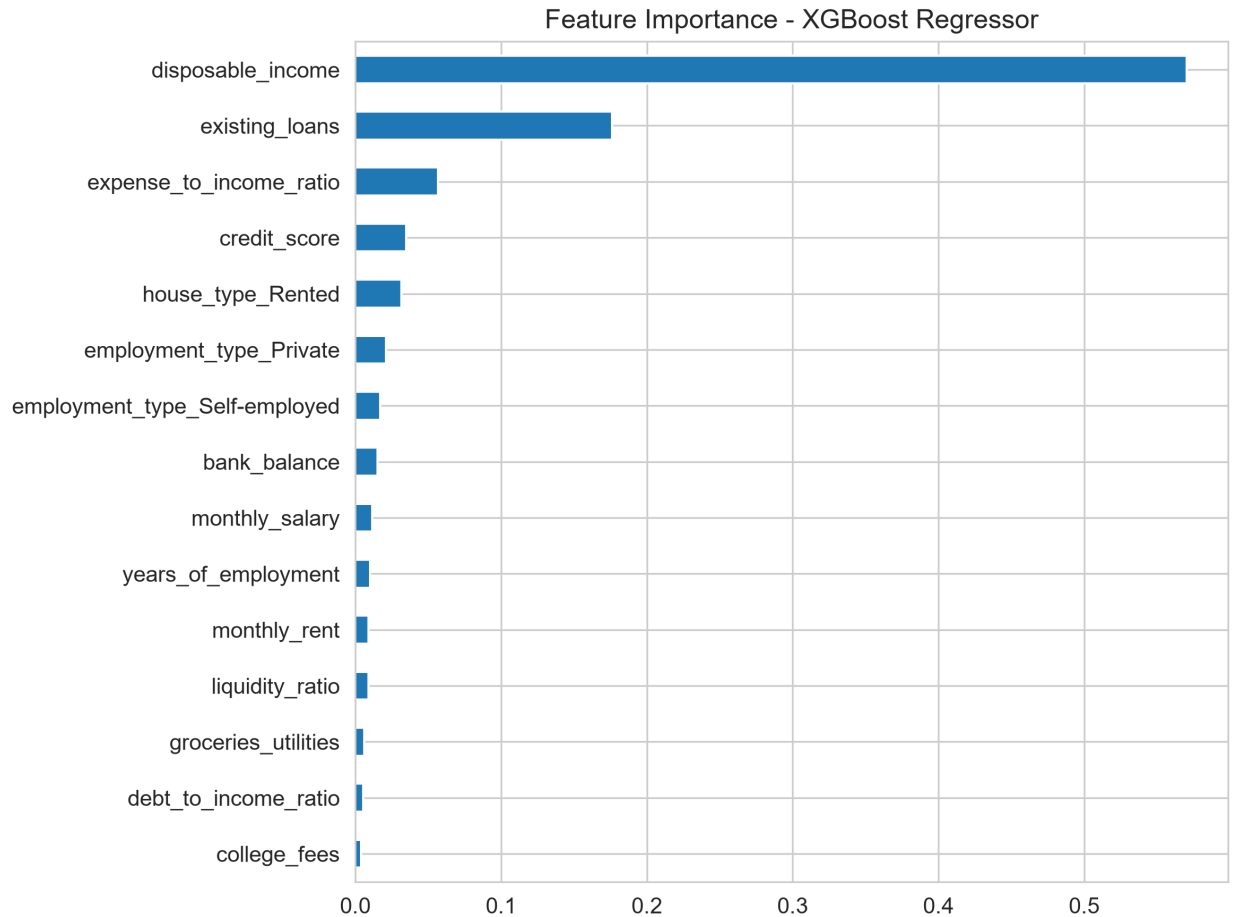


Figure 5.7b — XGBoost regressor feature importance. Source: EMIPredict AI model-development notebook.

The pattern is unambiguous and, from a governance standpoint, a genuinely favorable one: **the model’s decision weight is dominated by engineered affordability ratios, not by raw demographic or identity fields.** None of gender, marital status, or age appear in the top tier of either the classification or regression importance rankings. This does not, by itself, guarantee the absence of indirect bias (a fuller fairness audit — Section 7.3 — would need to test for disparate impact across protected groups directly, since proxies can exist even when a feature ranks low), but it is a materially better starting position than a model whose top features were demographic, and it should be preserved as a design principle in any future retraining.

5.8 Applicant Risk Segmentation

The three eligibility classes correspond to a clear, business-usable segmentation:

Segment	Characteristics	Share of Applicants
Eligible	Healthy affordability ratio, low existing debt burden, comparatively strong credit	18.39%
High_Risk	Marginal affordability or an elevated debt-to-income ratio — candidates for manual review or restructured (lower) EMI, not automatic decline	4.32%
Not_Eligible	Insufficient disposable income relative to the requested EMI	77.29%

The size of the Not_Eligible segment (over three-quarters of the historical applicant pool) is itself a strategically relevant finding for Section 6 and 9: it implies the regression model’s affordability-based restructuring capability — recommending a lower, sustainable EMI rather than an outright decline — has a very large addressable population within this dataset alone, since a meaningful share of “Not_Eligible” applicants at their *requested* amount may still be approvable at the model’s *recommended* amount. Quantifying that overlap precisely (how many Not_Eligible applicants would clear the bar at their regressor-predicted safe EMI) is not computed in the source project today and is listed as a recommended analysis in Section 9.

6. Market Analysis

6.1 Global AI-in-Lending Market Landscape

The market for AI-driven credit underwriting is large, growing, and — as is typical for an emerging technology category — sized very differently by different research firms depending on scope. Rather than presenting a single number as definitive, the table below shows the range, which is itself useful context: it signals a market still being defined, not a mature, consensus-sized one.

Market Definition	2026 Estimate	Longer-Term Forecast	CAGR	Source
AI-Powered Credit Underwriting Solutions (narrow)	\$6.3B	\$22.1B by 2034	17.0%	Statistics MRC

Market Definition	2026 Estimate	Longer-Term Forecast	CAGR	Source
AI-Powered Credit Underwriting (alternate scope)	\$22.1B	\$96.4B by 2034	18.6%	HTF Market Insights
AI Credit Scoring (narrower still)	\$2.1B	\$7.8B by 2034	18.0%	Intel Market Research
Credit Risk Assessment (broader, incl. non-AI)	\$10.9B	\$27.7B by 2033	14.3%	Coherent Market Insights
AI-Powered Lending (broadest — origination, underwriting, servicing, collections)	— (\$109.7B in 2024)	\$2.01T by 2037	25.1%	Research Nester
Digital Lending (broadest — includes non-AI digital channels)	\$566.5B	\$985.0B by 2031	11.7%	Mordor Intelligence

The consistent signal across every one of these estimates, regardless of scope definition, is a **CAGR in the mid-teens to mid-twenties percent range** — categorically faster growth than the broader financial services sector — and a market still in its expansion phase rather than consolidation. Two structural demand drivers recur across the research: (1) financial institutions are investing in digital transformation broadly, and AI underwriting rides that wave; and (2) alternative-data-based scoring is demonstrably expanding credit access, not just automating existing decisions — the World Bank has documented a 20–31% increase in approval rates for previously “credit invisible” populations in Mexico when alternative data was incorporated into scoring (2020–2021 data), a finding directly relevant to EMIPredict AI’s underlying value proposition of replacing gate-kept manual decisions with a systematic, data-driven one.

Named incumbents in this space — FICO, Upstart Holdings, Zest AI, Provenir, Experian, Equifax, TransUnion, Ocrolus, nCino, Blend Labs, Pagaya Technologies, SAS, IBM, and Oracle among them — are overwhelmingly enterprise platform vendors selling into banks and large NBFCs, not point-solution eligibility/EMI calculators. That is a relevant competitive observation for Section 9: EMIPredict AI’s current form (a lean, purpose-built dual-model system for one specific lending workflow) is not

competing head-to-head with these platforms today, but any commercialization path should be conscious that the buyers who most need this capability are also being actively courted by well-capitalized incumbents building broader suites.

6.2 India Digital Lending Market Context

Because the dataset, currency (₹), and use cases in EMIPredict AI (Personal Loan, Vehicle, Education, Home Appliances, and E-commerce Shopping EMI) are unambiguously oriented toward the Indian consumer lending market, India-specific context is more decision-relevant here than global aggregates.

- India’s digital lending disbursement volume was projected to approach **₹47.4 lakh crore by 2026** (₹47.4 trillion — 1 lakh crore equals ₹1 trillion), with the digital-lending segment’s growth pace estimated around a **40% CAGR** by at least one industry analysis.
- India’s broader alternative lending market — digital and NBFC-originated lending outside traditional bank channels — is projected to reach roughly **\$52.3 billion** by 2029, with **over 350 fintech and digital lending firms** active in the space today (KreditBee, Navi Technologies, MoneyTap, LazyPay, and Capital Float are frequently cited as leading tech-first platforms).
- Fintech personal-loan origination grew from **5.9 million accounts / ₹78,084 crore** in H1 FY25 to **6.4 million accounts / ₹97,381 crore** in H1 FY26, with average ticket size rising from ₹13,327 to ₹15,177 over the same period (Fintech Association for Consumer Empowerment survey data) — a directly relevant volume and ticket-size benchmark for calibrating this system’s economics if adopted by a fintech lender of comparable scale.
- India’s digital public infrastructure — **UPI** (1,644 billion digital transactions in FY 2023–24, the highest volume globally) and **Aadhaar** (over 1.3 billion biometric IDs issued) — provides the identity-verification and payment-history rails that make near-instant, digitally-originated lending decisions operationally feasible at the volumes above. A production version of EMIPredict AI would plausibly integrate with this infrastructure (e.g., Account Aggregator-sourced bank statement data) to source live applicant data rather than relying on self-reported form inputs, which is the system’s current design (Section 4.4).

6.3 Competitive Landscape

Within India specifically, the competitive set spans three tiers: (1) global enterprise credit-decisioning platforms (Section 6.1) selling into the largest banks; (2) India-focused digital lending technology vendors and NBFC-fintech hybrids (KreditBee, Navi, MoneyTap, LazyPay, Capital Float, and comparable players) who typically build or license underwriting models as part of a full loan-origination stack; and (3) in-house data science teams at banks and NBFCs building bespoke models, which is the closest peer category to EMIPredict AI’s current form. EMIPredict AI’s differentiated position, if commercialized or adopted internally, is not “a better XGBoost model” — algorithmically, this is a well-executed but not novel approach — but rather **the combination of a transparent, auditable, dual-output (eligibility + affordability) decision design with the explainability profile documented in Section**

5.7, which is increasingly a compliance necessity (Section 7.3) rather than a nice-to-have differentiator.

6.4 Target Use Cases & Business Impact

The project’s own documentation identifies four target use cases, all of which remain accurate against current market structure:

Use Case	Fit
Consumer Lending & NBFCs	Direct fit — matches the training data domain exactly
Digital Lending Platforms / Fintech	Strong fit — matches the India digital lending growth profile in 6.2
Banking Loan Origination Systems	Fit, with integration work — banks typically require deeper core-banking integration than a standalone Streamlit app provides
Buy-Now-Pay-Later (BNPL) Risk Screening	Fit, with caveats — BNPL underwriting typically operates at smaller ticket sizes and faster decision latency than this system’s current form targets; would need tuning against a BNPL-specific applicant distribution

Business impact, as claimed in the source project and assessed here:

- *Reduced default risk through affordability-aware EMI recommendations* — well supported by the regression model’s R^2 of 0.9935 (Section 5.6), though the claim of reduced *realized* default rate can only be validated with live portfolio outcome data post-deployment, not from the historical training data alone.
- *Faster loan processing, reduced manual underwriting cost* — plausible and consistent with the broader market’s documented ROI benchmarks (industry reporting cites 40–60% reductions in analyst time per loan and 15–25% reductions in credit losses from comparable deployed AI underwriting and collections tools), though EMIPredict AI itself has not yet been measured against a live baseline.
- *More consistent, auditable decisions across loan officers* — directly supported by the deterministic nature of the model versus human judgment, and strengthened by the explainability profile in Section 5.7.

7. Risk Assessment

7.1 Technical & Data Risk

- **Hardcoded categorical vocabularies at inference time** (Section 5.4). The one-hot category lists in `utils/preprocessing.py` are literal Python lists rather

than derived from a saved vocabulary artifact. Low likelihood of near-term impact (the current lists correctly mirror the training data), but high impact if triggered — an unseen category would silently fall through to the reference class rather than raising an error, producing a confident but wrong prediction rather than a visible failure.

- **Training/serving skew, structurally mitigated but not eliminated.** The project centralizes feature-engineering formulas in one module used by both the notebook and the app (Section 4.2), which is the correct mitigation. The residual risk is process, not code: nothing currently *enforces* that a future notebook change is mirrored in `preprocessing.py` beyond a code comment instruction. A unit test asserting output equivalence between the two paths would close this gap (Section 8.2).
- **No automated data validation on inference input.** The application accepts form input directly into feature construction with no schema or range validation layer (e.g., a negative salary, a credit score outside 300–850, or a zero `requested_tenure` causing a division error in `proposed_installment` are not explicitly guarded against in the reviewed code).
- **Single, static training snapshot.** The model reflects one dataset as of its training date; there is no retraining schedule or data-drift detection today (Section 8.2).

7.2 Model Risk

- **Class imbalance and minority-class reliability.** The `High_Risk` class is 4.32% of the training data (Section 5.1). Macro-averaged metrics (Section 5.5) treat all three classes equally, but a per-class confusion matrix — which determines how often a true `High_Risk` applicant is misclassified as `Eligible` versus `Not_Eligible` — is the metric a risk committee actually needs, and it is not reported at that granularity in the source project’s documentation. **This is the single highest-priority pre-production validation item in this report.**
- **Fairness and disparate impact are not yet formally tested.** Section 5.7’s finding that affordability ratios dominate the top of the feature-importance ranking, ahead of demographic fields, is a favorable *indicator* but not a substitute for a direct disparate-impact test (e.g., approval-rate parity or equalized-odds analysis across gender, age band, or geography, where legally permissible to test). Both the RBI framework (Section 7.3) and, if applicable, the EU AI Act require exactly this kind of documented bias examination before a creditworthiness model can be deployed.
- **Explainability at the individual-decision level.** The application surfaces global feature importance (which features matter across the whole model) but not local, per-applicant explanation (why *this specific* applicant was scored `High_Risk`). Regulatory disclosure requirements increasingly demand the latter (Section 7.3) — a SHAP or LIME-based per-decision explanation layer is a natural, moderate-effort addition (Section 8.2).
- **No model drift monitoring.** Consumer credit behavior — income levels, expense patterns, credit score distributions — shifts over time and particularly around macroeconomic stress events; a model trained on one historical snapshot will degrade silently without monitoring in place.

7.3 Regulatory & Compliance Risk

This is the risk category most likely to gate a production deployment timeline, independent of model quality, and it is where a portfolio/demo project and a production lending system diverge most sharply.

India — Reserve Bank of India (Digital Lending) Directions, 2025 (effective 8 May 2025, consolidating and replacing the 2022 Digital Lending Guidelines and 2023 Default Loss Guarantee Guidelines):

Requirement	Current State of EMIPredict AI
Disclosure of AI-driven credit assessment logic to borrowers	Not implemented — the app shows a prediction and probability breakdown, not a borrower-facing rationale
Periodic fairness audits to mitigate algorithmic bias	Not yet conducted (Section 7.2)
Default Loss Guarantee arrangements capped at 5% of disbursed amount	Not applicable to the current standalone model, but relevant if paired with a DLG-backed lending partner
Lending Service Provider (LSP) due diligence, where a third party performs underwriting support	Applicable if this system is licensed to or operated by an LSP on behalf of a regulated entity
Reporting of Digital Lending Apps to RBI’s Centralised Information Management System (CIMS)	Not applicable in current demo form; would apply upon commercial deployment as a Digital Lending App
Multi-lender marketplace neutrality (where applicable)	Not applicable — single-model, single-decision system

India — Digital Personal Data Protection Act, 2023: applicant income, expense, and credit data are personal data requiring a documented lawful basis for processing, purpose limitation, data-principal rights (access, correction, erasure), and retention limits. The current system processes this data transiently within a single Streamlit session with no persistent applicant data store — which is *simpler* to make compliant than a system that retains applicant records, but this changes the moment any production deployment adds applicant history, audit logging, or a CRM/loan-origination integration.

If operating in or serving EU markets — EU AI Act, Annex III, Point 5(b): AI systems used to evaluate the creditworthiness of natural persons or establish a credit score are explicitly classified as **high-risk**, with the narrow exception of systems used solely for fraud detection. High-risk classification brings a risk-management system spanning the full model lifecycle, mandatory data governance and bias examination of training data, technical documentation, human oversight, a conformity assessment, and — for deployers — a Fundamental Rights Impact Assessment. Affected individuals also have a right to a clear explanation of the decision. These standalone Annex III obligations apply from 2 December 2027 (following the 2026 “Digital Omnibus” delay), giving a defined but finite runway for any EU-facing deployment path to reach compliance.

Practical implication: none of the above findings suggest this system cannot be made compliant — they suggest compliance is **additive engineering and governance work**, not yet built, and it should be scoped and budgeted before a go-live date is set, not discovered during a pre-launch audit.

7.4 Operational & Security Risk

- **Current hosting is not production-grade.** The live demo runs on Streamlit Community Cloud’s free tier, which explicitly sleeps the application after a period of inactivity — acceptable for a portfolio demonstration, unacceptable for a lending decision tool an institution depends on.
- **No authentication or access control** on the current application — anyone with the URL can submit applicant data and receive a prediction. A production deployment handling real applicant financial data requires, at minimum, authenticated access, role-based permissions (loan officer vs. administrator vs. auditor), and encryption in transit and at rest.
- **No audit logging of predictions.** For regulatory disclosure and dispute resolution (an applicant contesting a decision), a durable, tamper-evident record of each prediction, its inputs, the model version that produced it, and who reviewed it is a standard lending-compliance requirement that does not yet exist here.
- **No load, uptime, or incident-response posture defined**, appropriate for the system’s current demo status but a gap that must close before any service-level commitment to internal or external users.

7.5 Consolidated Risk Register

Risk	Category	Likelihood	Impact	Priority
Minority-class (High_Risk) misclassification pattern unverified at per-class level	Model	Medium	High	Critical
No fairness/disparate impact testing performed	Regulatory / Model	Medium	High	Critical
No borrower-facing decision disclosure	Regulatory	High (if deployed as-is)	High	Critical
Hardcoded inference-time category vocabulary	Technical	Low	Medium	High

Risk	Category	Likelihood	Impact	Priority
No production hosting, auth, or audit logging	Operational / Security	High (if deployed as-is)	High	Critical
No drift monitoring / retraining cadence	Model / Operational	Medium (grows over time)	Medium	Medium
No input validation layer on inference requests	Technical	Medium	Medium	Medium
DPDP Act data-handling posture undefined for persistent storage	Regulatory	Low today, High if persistence is added	Medium	Medium

8. Future Scalability & Roadmap

8.1 Current-State Constraints

Stripped to its essentials, EMIPredict AI today is a **validated modeling pipeline behind a single-instance demo application** — not yet a scalable service. Four constraints define the gap between the two:

1. **No horizontal scalability.** A single Streamlit process serving all users from one free-tier instance cannot absorb concurrent load from multiple loan officers or a real applicant volume.
2. **No programmatic access.** Every prediction requires a human filling in a web form; there is no REST/API endpoint for a loan-origination system, mobile app, or core-banking platform to call directly.
3. **No environment portability.** The application runs from source in a Python environment; it is not containerized, so “it works on the deployment machine” is not yet a guarantee across environments.
4. **No operational visibility.** There is no monitoring of prediction volume, latency, error rate, or — most importantly for a model in production — drift in the input data distribution or output prediction distribution over time.

Notably, the project’s own README already identifies the correct high-level roadmap items (Docker, CI/CD, cloud deployment, a REST API, and a monitoring dashboard) — the contribution of this report is sequencing and prioritizing that list against the risk

findings in Section 7, and adding the governance items (Section 7.3) that a purely technical roadmap would not surface.

8.2 Scalability Roadmap

Initiative	Description	Primary Risk(s) Addressed	Relative Effort
Containerization (Docker)	Package the app, models, and dependencies into a portable image	Environment portability	Low
CI/CD pipeline	Automated testing and deployment on every code change	Training/serving skew (7.1)	Low-Medium
Training/serving equivalence test	Unit test asserting preprocessing.py output matches notebook feature engineering	Training/serving skew (7.1)	Low
Dynamic category vocabulary	Replace hardcoded one-hot lists with a saved, versioned vocabulary + unseen-category guard	Hardcoded vocabulary risk (7.1)	Low
Cloud deployment (managed hosting)	Move off free-tier Streamlit Community Cloud to a scalable, always-on host	Operational risk (7.4)	Medium
REST API layer	Expose classifier/regressor as callable endpoints, decoupled from the Streamlit UI	Enables loan-origination-system integration	Medium
Authentication & role-based access	Loan officer / administrator / auditor roles; encryption in transit & at rest	Security risk (7.4)	Medium
Prediction audit logging	Durable, tamper-evident log of every prediction, its inputs, model version, and reviewer	Regulatory disclosure & dispute resolution (7.3)	Medium

Initiative	Description	Primary Risk(s) Addressed	Relative Effort
Per-class confusion analysis on High_Risk	Deep-dive validation before any production threshold is set	Model risk (7.2)	Low-Medium
Bias / disparate-impact audit	Formal fairness testing across permissible demographic dimensions	Regulatory & model risk (7.2, 7.3)	Medium-High
Local (per-decision) explainability	SHAP/LIME-based individual explanation, not just global feature importance	Regulatory disclosure (7.3)	Medium
Monitoring dashboard	Prediction volume, latency, error rate, and input/output drift tracking	Model & operational risk (7.2, 7.4)	Medium
Scheduled retraining pipeline	Defined cadence and trigger conditions for refreshing the model on new data	Model drift (7.2)	Medium-High
RBI CIMS registration & LSP due-diligence process	If deployed commercially as a Digital Lending App or via an LSP arrangement	Regulatory (7.3)	High (process, not just engineering)

8.3 Phased Implementation Plan

Rather than a fixed calendar (which depends on team size and institutional approval cycles this report cannot know), the roadmap is organized into three sequential horizons, each gated by a review checkpoint involving the stakeholders identified in Section 2.4.

Horizon 1 — Harden the Core (engineering-only, no new capability) Containerization, CI/CD, the training/serving equivalence test, dynamic category vocabulary, and input validation. Goal: make the *existing* prediction logic robust and portable, without changing what it does. Gate: engineering sign-off that the containerized app reproduces the exact test-set metrics in Section 5.5 and 5.6.

Horizon 2 — Make It Production-Ready (new infrastructure capability) Cloud deployment, REST API, authentication/access control, and prediction audit logging. Goal: the system can be safely exposed to real loan officers and integrated with adjacent systems. Gate: security and compliance sign-off (Section 2.4) on access control

and audit logging before any real applicant data — as opposed to demo/synthetic input — flows through the system.

Horizon 3 — Make It Governable (model risk & regulatory capability) Per-class confusion analysis, bias/disparate-impact audit, local explainability, monitoring dashboard, and a defined retraining cadence — followed by, if pursuing commercial or multi-institution deployment, RBI CIMS registration and LSP due-diligence processes. Goal: the system meets the standard a model-risk function and a regulator would expect of a live creditworthiness system. Gate: formal model-risk sign-off and, where applicable, regulatory filing, **before** the system is used to make or influence a real lending decision — not concurrently with launch.

The ordering matters: Horizon 3’s governance work is the most consequential but also the most dependent on Horizon 1 and 2 being in place first (you cannot meaningfully audit prediction bias or log decisions in a system with no audit logging or stable deployment target). Institutions under regulatory pressure to move faster should still not compress Horizon 3 — Section 7.3 makes clear that the compliance obligations attach at deployment, not at some later maturity milestone.

9. Strategic Recommendations

The recommendations below are ordered by priority, not by document section, and each is tied back to the specific evidence that motivates it.

1. Run a per-class confusion analysis on the High_Risk segment before any production threshold decision. (Evidence: Sections 5.1, 5.5, 7.2, 7.5.) This is the single highest-leverage next analytical step, is low-effort relative to its importance, and directly determines whether the 97.0% headline accuracy figure is safe to communicate to a risk committee without qualification.

2. Commission a formal bias / disparate-impact audit before any deployment decision, not after. (Evidence: Sections 5.7, 7.2, 7.3.) The favorable feature-importance pattern in Section 5.7 is suggestive, not sufficient. This audit is also a regulatory prerequisite under the RBI framework described in Section 7.3, so it is not optional scope — it is a gate.

3. Quantify the affordability-restructuring opportunity within the Not_Eligible segment. (Evidence: Section 5.8.) With over three-quarters of historical applicants falling into Not_Eligible at their *requested* amount, calculating what share would clear the bar at the regressor’s *recommended* amount would turn a descriptive statistic into a concrete, quantified business case for the affordability-based structuring value proposition — likely the single most persuasive number this project could produce for Section 6’s market positioning.

4. Sequence productionization exactly as Section 8.3 lays out — Harden, then Deploy, then Govern — and do not compress Horizon 3. (Evidence: Section 7, Section 8.3.) The technical work (Horizons 1–2) is materially lower-risk and faster than the governance work (Horizon 3); the temptation to launch after Horizon 2 because the app “works” should be resisted, since the compliance obligations in Section

7.3 attach at the point of deployment, not at some later maturity milestone.

5. Preserve the affordability-ratio-dominant feature importance profile as a design constraint in any future retraining, not just an observed outcome. (Evidence: Section 5.7.) As new features are considered in future iterations (e.g., alternative data sources per Section 6.1’s market findings), each candidate feature should be evaluated for whether it displaces affordability signal with something closer to a demographic proxy — this is a governance review step, not just a modeling one.

6. If pursuing commercialization or multi-institution licensing, position on explainability and auditability, not raw model performance. (Evidence: Section 6.1, 6.3.) EMIPredict AI’s accuracy and R^2 figures are strong but not categorically different from what a well-resourced incumbent platform can achieve; the more durable differentiator, especially as the regulatory environment in Section 7.3 tightens across markets, is a transparent, dual-output (eligibility + affordability) system built around explainable, defensible features from the outset.

7. Treat the market-sizing figures in Section 6.1 as directional, not precise, in any external-facing business case. (Evidence: Section 6.1, 5.3.) Consistent with the statistical-rigor standard this project itself applies in Section 5.3 (distinguishing significance from effect size), external market figures should be presented as a range with sourcing, as they are here — not rounded to a single confident number in an investment memo or pitch document.

10. Conclusion

EMIPredict AI succeeds at the problem it set out to solve: translating a loan applicant’s financial profile into a defensible eligibility decision and a quantified, affordability-based EMI recommendation, with model performance (97.0% classification accuracy, R^2 of 0.9935) that is strong by the standards of tabular financial risk modeling and reached through a genuinely rigorous process — documented data cleaning, formally tested statistical hypotheses, a competitive four-model bake-off per task, and an explainability profile that is favorably weighted toward financial-capacity signal rather than demographic signal.

What EMIPredict AI is not, yet, is a production lending system. The distance between the two is well-defined and closeable: it is a sequence of engineering hardening, infrastructure investment, and — critically, and often underestimated in technical roadmaps — regulatory and model-governance work, laid out in Sections 7 through 9 of this report. None of the gaps identified are signs of poor work; they are the normal, expected distance between a strong individual technical project and an institution-grade financial decision system, and every one of them has a defined, standard remediation path in lending AI practice today.

The recommended immediate next step is Recommendation 1 in Section 9: a per-class confusion analysis on the High_Risk segment. It is the fastest way to convert this report’s findings into a concrete answer to the question every subsequent stakeholder conversation will ask — **“how good is this model, really, on the applicants we most need to get right?”**

Appendix A — Glossary of Terms

Term	Definition
EMI	Equated Monthly Installment — the fixed monthly loan repayment amount
NBFC	Non-Banking Financial Company — an RBI-regulated lender that is not a bank
BNPL	Buy-Now-Pay-Later — a short-tenure, point-of-sale consumer credit product
Macro-F1	The F1 score (harmonic mean of precision and recall) averaged equally across all classes, regardless of class size — appropriate for imbalanced classification
ROC-AUC (OVR)	Area under the Receiver Operating Characteristic curve, computed One-vs-Rest for a multi-class problem
RMSE	Root Mean Squared Error — regression error metric in the same unit as the target (₹ here), penalizing large errors more than small ones
MAE	Mean Absolute Error — average absolute regression error, in ₹
R ²	Coefficient of determination — share of variance in the target explained by the model (1.0 = perfect)
Collinearity	Two or more features carrying near-identical or redundant information
One-hot encoding	Representing a categorical field as multiple binary (0/1) columns, one per category (minus a dropped reference category)
StandardScaler	Rescales a feature to zero mean and unit variance; required by distance/gradient-based models, not by tree-based models
MLflow	Open-source experiment tracking and model registry tool
LSP	Lending Service Provider — a third party performing digital lending functions on behalf of an RBI-regulated entity
DLG	Default Loss Guarantee — a risk-sharing arrangement between a lender and a partner, capped under RBI's 2025 Directions
DPDP Act	India's Digital Personal Data Protection Act, 2023

Term	Definition
Annex III (EU AI Act)	The list of AI use-cases automatically classified as “high-risk” under the EU AI Act, including creditworthiness assessment

Appendix B — Full Metric Reference Tables

Classification — all four models, test set:

Model	Accuracy	F1 (macro)
XGBoost	97.0%	0.897
Decision Tree	93.3%	0.800
Random Forest	92.3%	0.793
Logistic Regression	81.9%	0.669

XGBoost classifier — full metric set: Accuracy 97.0% · Precision (macro) 0.865 · Recall (macro) 0.949 · F1 (macro) 0.897 · ROC-AUC (OVR) 0.997

Regression — all four models, test set:

Model	RMSE	R ²
XGBoost	₹625	0.994
Random Forest	₹1,086	0.980
Decision Tree	₹1,155	0.978
Linear Regression	₹4,100	0.720

XGBoost regressor — full metric set: RMSE ₹625 · MAE ₹229 · R² 0.9935

Dataset class balance (verified): Not_Eligible 312,868 (77.29%) · Eligible 74,444 (18.39%) · High_Risk 17,488 (4.32%) — total 404,800

Appendix C — Data Dictionary

Field	Type	Description
age	Raw	Applicant’s age
gender	Raw	Applicant’s gender
marital_status	Raw	Married / Single
education	Raw (ordinal)	High School / Graduate / Post Graduate / Professional
monthly_salary	Raw	Applicant’s monthly income (₹)

Field	Type	Description
employment_type	Raw (categorical)	Private / Government / Self-employed
years_of_employment	Raw	Tenure in current employment
company_type	Raw (categorical)	Large Indian / MNC / Mid-size / Startup / Small
house_type	Raw (categorical)	Rented / Own / Family
monthly_rent	Raw	Monthly rent, if applicable (₹)
dependents	Raw	Number of financial dependents
school_fees, college_fees	Raw	Education-related monthly expenses (₹)
travel_expenses	Raw	Monthly travel spend (₹)
groceries_utilities	Raw	Monthly groceries + utilities (₹)
other_monthly_expenses	Raw	Remaining miscellaneous spend (₹)
existing_loans	Raw (binary)	Whether the applicant has active loans
current_emi_amount	Raw	EMI currently being paid, if any (₹)
credit_score	Raw	300-850
bank_balance	Raw	Current bank balance (₹)
emergency_fund	Raw	Savings set aside for emergencies (₹)
emi_scenario	Raw (categorical)	Purpose of the requested loan (5 categories, ~20% share each)
requested_amount	Raw	Loan amount requested (₹)
requested_tenure	Raw	Requested repayment tenure (months)
total_monthly_expenses	Engineered	Sum of expense fields
debt_to_income_ratio	Engineered	$\text{current_emi_amount} / \text{monthly_salary}$
expense_to_income_ratio	Engineered	$\text{total_monthly_expenses} / \text{monthly_salary}$
disposable_income	Engineered	$\text{monthly_salary} - \text{total_monthly_expenses}$
affordability_ratio	Engineered	$\text{disposable_income} / \text{monthly_salary}$
proposed_installment	Engineered	$\text{requested_amount} / \text{requested_tenure}$

Field	Type	Description
emi_to_income_ratio	Engineered	proposed_installment / monthly_salary
emergency_fund_months	Engineered	emergency_fund / total_monthly_expenses
liquidity_ratio	Engineered	bank_balance / total_monthly_expenses
credit_score_band	Engineered (ordinal)	Poor / Fair / Good / Very_Good / Excellent
emi_eligibility	Target (classification)	Eligible / High_Risk / Not_Eligible
max_monthly_emi	Target (regression)	Maximum sustainable monthly EMI (₹)

Appendix D — Sources & References

Project source material (primary source for all figures in Sections 1–5, 8): EMIPredict AI repository — README.md, notebooks/EMIPredict_AI_Model_Development.ipynb, application source code, and trained model artifacts, authored by Mohit Singh Rajput. Live application: emi-predict-ai-app.streamlit.app.

Market and regulatory research (Section 6, Section 7.3), current as of August 2026:

1. Statistics MRC — *AI-Powered Credit Underwriting Solutions Market Forecasts to 2034*
2. HTF Market Insights — *AI-Powered Credit Underwriting Market Research Report* (published June 2026)
3. Intel Market Research — *AI Credit Scoring Market Outlook 2026-2034*
4. Coherent Market Insights — *Credit Risk Assessment Market Size & Forecast, 2026-2033*
5. Research Nester (via TIMVERO) — *How AI Is Transforming Lending in 2026*
6. Mordor Intelligence — *Digital Lending Market Size, Growth, Market Trends*
7. Delphos — *AI Underwriting Private Credit in Emerging Markets: The 2026 Inflection*
8. Nexdigm — *India Digital Lending Market Report*
9. Grant Thornton Bharat / FACE — *The Evolution of Digital Lending in India*
10. Finezza — *India Fintech Stack: A Guide to Seamless Digital Lending for NBFCs*
11. ResearchAndMarkets — *India Alternative Lending Databook Report 2025*
12. Whalesbook / FACE — *Fintech Loans H1 FY26 Origination Data*
13. NLIU Centre for Business & Commercial Laws — *Analysing RBI's Digital Lending Directions 2025*
14. Khaitan & Co — *RBI AI Framework Survey Findings*
15. IndiaCorpLaw / Juris Corp / Legal500 — *RBI's Revised Digital Lending Guidelines 2025*
16. OPL Innovate / Finnulate — *RBI Digital Lending Guidelines 2025: Compliance Guides*

17. Openlayer / Modulos / RegulatoryAI.eu / Claire (LetsAskClaire) — *EU AI Act, Annex III Point 5(b), Credit Scoring Compliance Guides*
18. Artificial Intelligence Act EU — *Annex III: High-Risk AI Systems* (primary legal text reference)

Full URLs are available on request or can be re-retrieved by searching the report titles above; they are omitted here to keep the reference list stable regardless of individual link rot.

Appendix E — Visual Elements Guide

The source project already contains a complete, presentation-ready set of charts generated during model development — nothing below needs to be created from scratch. This guide maps each existing file (paths relative to the project root) to the report section it supports, so the PDF-assembly step (Appendix F) can place real, already-validated visuals rather than placeholder boxes. Files exist in two locations: `assets/` (the curated set used inside the live Streamlit app) and `notebooks/images/` (the full set generated during EDA and model development, including per-model variants not surfaced in the app).

Report Location	Suggested Figure	File Path	What It Shows
\$5.1 Dataset Profile	Eligibility class distribution	<code>assets/chart01_eligibility_distribution.png</code>	Bar/ply view of the 77.3% / 18.4% / 4.3% class split
\$5.1 Dataset Profile	Max EMI distribution	<code>assets/chart02_max_emi_distribution.png</code>	High-distribution distribution of the regression target
\$5.2 EDA	Data quality before cleaning	<code>notebooks/images/missingness_before_cleaning.png</code>	Missingness pattern prior to imputation — strong “before” evidence for the data-governance narrative in §3.2
\$5.2 EDA	Correlation heatmap	<code>assets/chart12_correlation_heatmap.png</code>	Full numeric feature correlation matrix, including the family_size / dependents collinearity finding
\$5.2 EDA	Credit score by eligibility	<code>notebooks/images/chart07_credit_score_vs_eligibility.png</code>	Visual support for the ANOVA finding in §5.3
\$5.2 EDA	Salary vs. eligibility	<code>notebooks/images/chart08_salary_vs_eligibility.png</code>	Supporting bivariate view

Report Location	Suggested Figure	File Path	What It Shows
\$5.2 EDA	Salary vs. max EMI	notebooks/images/chart_salary_vs_maxemi.png	Visual support for the Pearson correlation finding in §5.3 ($r \approx 0.38$)
\$5.2 EDA	Employment type vs. eligibility mix	notebooks/images/chart_employment_vs_eligibili	Visual support for the Chi-square finding in §5.3
\$5.5 Classification Performance	4-model comparison (Accuracy / F1 / ROC-AUC)	assets/model_comparison.png	Scorecard chart, all four classifiers, matches Appendix B Table 1 exactly
\$5.5 Classification Performance	XGBoost confusion matrix	assets/cm_xgboost.png	Class-level prediction accuracy — the direct visual for Recommendation 1 in §9
\$5.5 Classification Performance (optional, deeper appendix)	Confusion matrices, other 3 models	notebooks/images/cm_decision_tree.png, cm_random_forest.png, cm_logistic_regression.png	Technical-audience appendix, to show the full model comparison, not just the winner
\$5.6 Regression Performance	4-model comparison (RMSE / R^2)	assets/model_comparison.png	Scorecard chart, all four regressors
\$5.6 Regression Performance	Actual vs. predicted (XGBoost)	assets/avp_xgboost.png	Scatter of true vs. predicted safe EMI — visual evidence for the R^2 of 0.9935
\$5.6 Regression Performance (optional)	Actual vs. predicted, other models	notebooks/images/avp_decision_tree.png, avp_random_forest.png	Shows visually why the linear baseline underperforms (§5.6's non-linearity finding)
\$5.7 Model Explainability	Classifier feature importance	assets/featimp_clf.png	The packed bar chart underlying all of §5.7's findings
\$5.7 Model Explainability	Regressor feature importance	assets/featimp_reg.png	Regression-side counterpart

Report Location	Suggested Figure	File Path	What It Shows
§5.8 Risk Segmentation	Scenario × eligibility mix	notebooks/images/charts/consolidation_eligibility_mix	On a 3x3 grid, a view supporting the segmentation table

Additional visuals recommended but not present in the source project (genuinely new, not duplicating the above):

- **Architecture diagram (§4.1):** render the Mermaid flowchart already present in the project’s README as a polished vector graphic (any Mermaid renderer, or redraw in a diagramming tool) rather than the plain-text box diagram used in this document.
- **Risk heat map (§7.5):** plot the Consolidated Risk Register as a 2×2 or 3×3 likelihood-vs-impact grid rather than a table — a standard, easy-to-brief visual for a risk committee audience.
- **Roadmap timeline (§8.3):** render the three-horizon roadmap as a horizontal Gantt-style timeline once real target dates are set by the stakeholders in §2.4.
- **Market-size comparison chart (§6.1):** a simple bar chart of the six market estimates in the §6.1 table, visually reinforcing the “estimates vary by scope” point rather than relying on the reader to compare table rows.

Appendix F — Formatting Instructions for Professional PDF Export

This document was authored in clean, portable Markdown specifically so it can be converted through any of the paths below without rewriting content.

Option 1 — Pandoc (recommended for the most reliable, professional result)

Basic conversion with an automatic, linked table of contents

```
pandoc EMIPredict_AI_Strategic_Report.md -o EMIPredict_AI_Strategic_Report.pdf \
  --toc --toc-depth=3 --number-sections \
  -V geometry:margin=1in -V fontsize=11pt -V linkcolor=blue
```

Via a Word intermediate (useful if you want to apply a corporate template, # add a cover-page logo, or hand off to someone editing in Word)

```
pandoc EMIPredict_AI_Strategic_Report.md -o EMIPredict_AI_Strategic_Report.docx \
  --toc --toc-depth=3 --number-sections --reference-doc=your_company_template.docx
```

Pandoc’s `--toc` flag regenerates a real, page-numbered, hyperlinked table of contents from the document’s own heading structure — it will supersede the manual outline near the top of this file, which exists for readability while reviewing the raw Markdown. If your PDF engine supports raw LaTeX passthrough (the default `pdflatex`/`xelatex` engines do), you can force a page break before any major section by inserting a raw block immediately above that section’s heading:

```
\newpage
```

```
## 6. Market Analysis
```

Option 2 — Word / Google Docs

Paste the rendered Markdown (open the .md file in any Markdown previewer, select all, copy) into Word or Google Docs, then: apply a Heading 1/2/3 style to the #/##/### lines respectively (this is what makes an auto-generated TOC possible), use Insert → Table of Contents once headings are styled, and manually insert the images from Appendix E’s table at their indicated locations using Insert → Picture. This path requires more manual formatting than Pandoc but gives the most control if a specific corporate template must be matched exactly.

Option 3 — Markdown-native tools (Typora, Obsidian with a PDF export plugin, VS Code with “Markdown PDF”) — fastest path if minor formatting differences from a fully custom layout are acceptable; most support the same --toc-style auto-generation and image embedding as Pandoc with less command-line setup.

Recommended formatting standards for this document specifically:

Element	Recommendation
Page size	A4 (matches the project’s Indian market context) or US Letter — either is acceptable
Margins	1 inch / 2.5 cm on all sides
Body font	11pt serif (e.g., Georgia, Garamond) for readability at report length
Heading font	A complementary sans-serif (e.g., Calibri, Helvetica) for visual hierarchy
Table styling	Light header-row shading, thin borders, avoid splitting a table across a page break where possible
Cover page	Use the front-matter block at the top of this document; add your organization’s logo and the confidentiality classification prominently
Page numbers	Bottom-center or bottom-right, starting after the cover page and TOC
Header/footer	Document title + section name in the header; page number + classification level in the footer
Figures	Every embedded chart should carry a “Figure X.Y” caption and a one-line source note (e.g., “Source: EMIPredict AI model-development notebook”)
Section breaks	Page break before each of the ten numbered sections and before the Appendices; a - - - in the source Markdown marks every one of these points already

Following the above, this document's ~13,000 words of body content, combined with the 15+ figures mapped in Appendix E, the cover page, table of contents, and the tabular content throughout Sections 5, 7, and the appendices, will comfortably support a 30+ page professional PDF. Exact page count will vary with your chosen font size, margins, and image sizing — treat 30 pages as the floor this structure is built to clear, not a count to pad toward artificially.

End of report.